

BFS Traversal

Step 1: Start at E

Legend

Complete

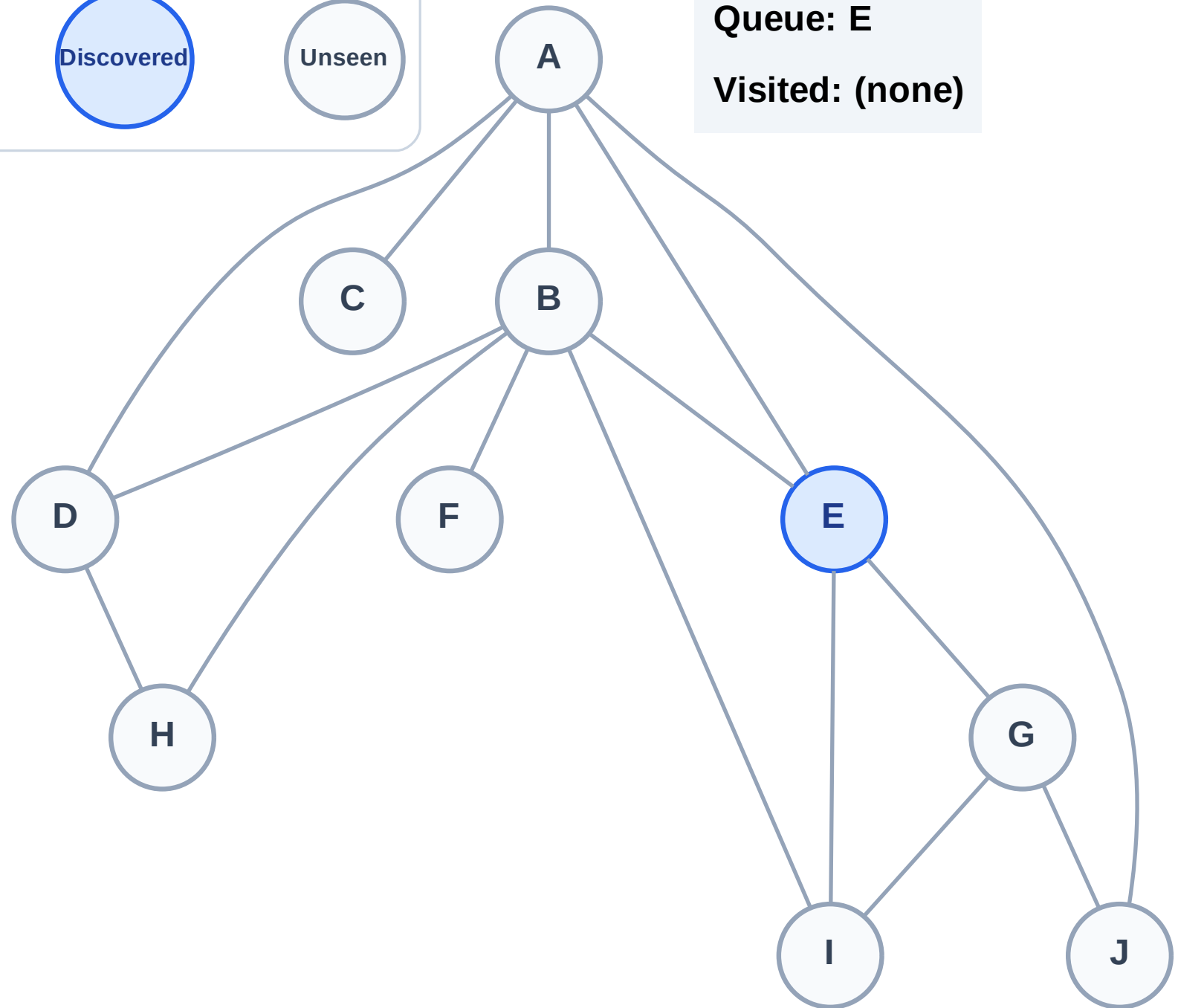
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

Complete

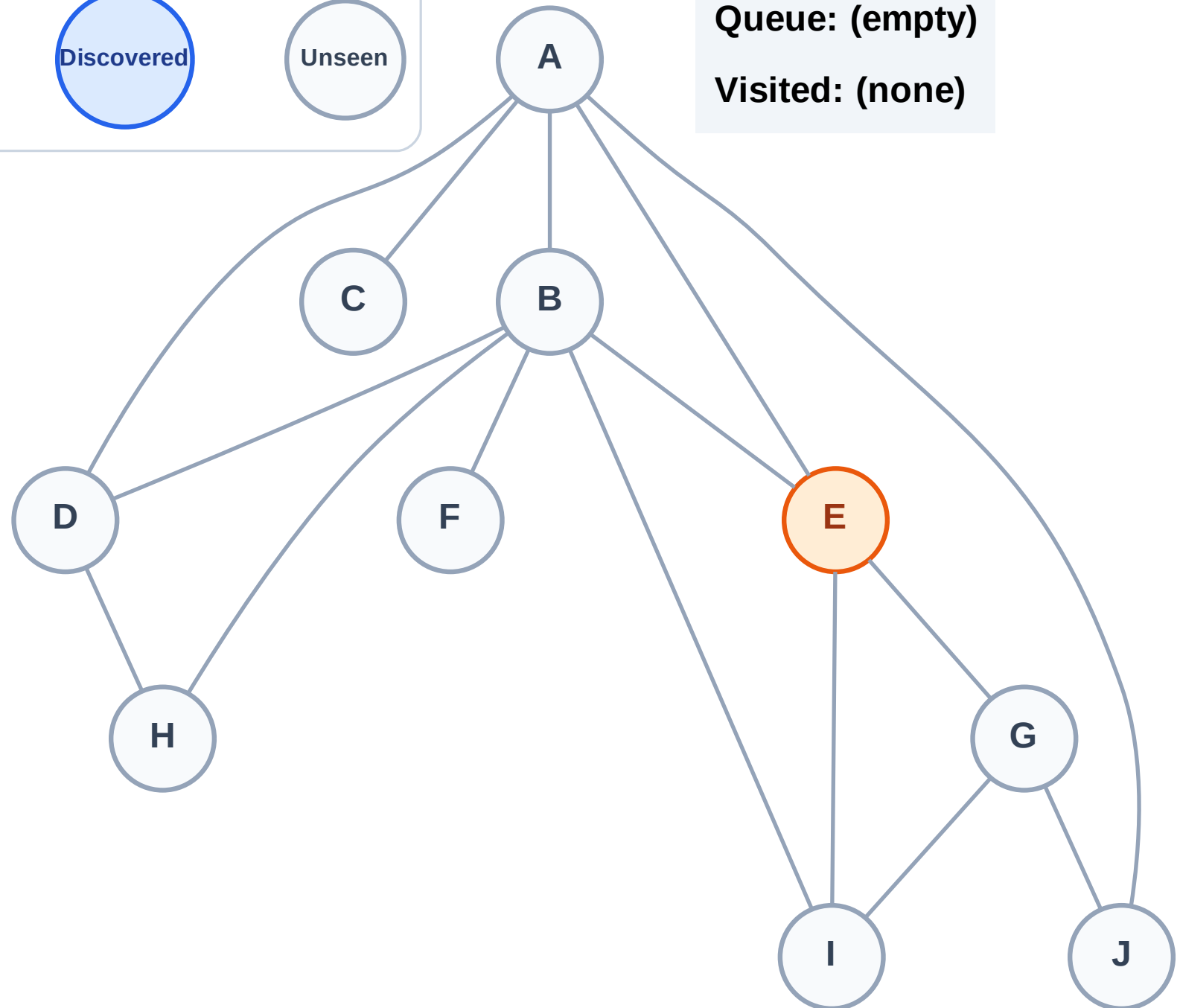
Processing

Discovered

Unseen

Queue: (empty)

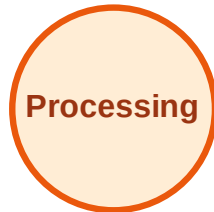
Visited: (none)



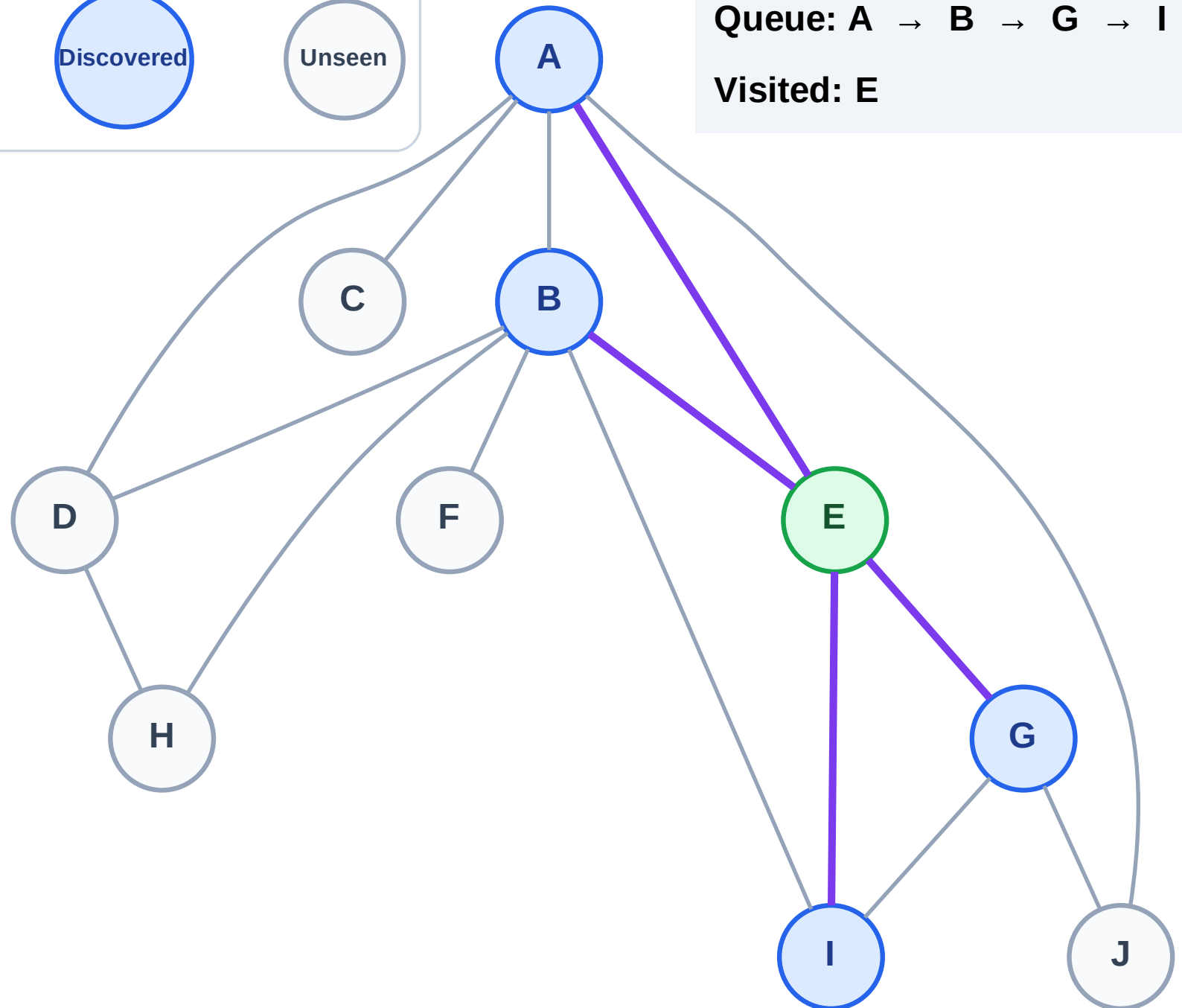
BFS Traversal

Step 3: Discover A, B, G, I from E

Legend



Queue: A → B → G → I
Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

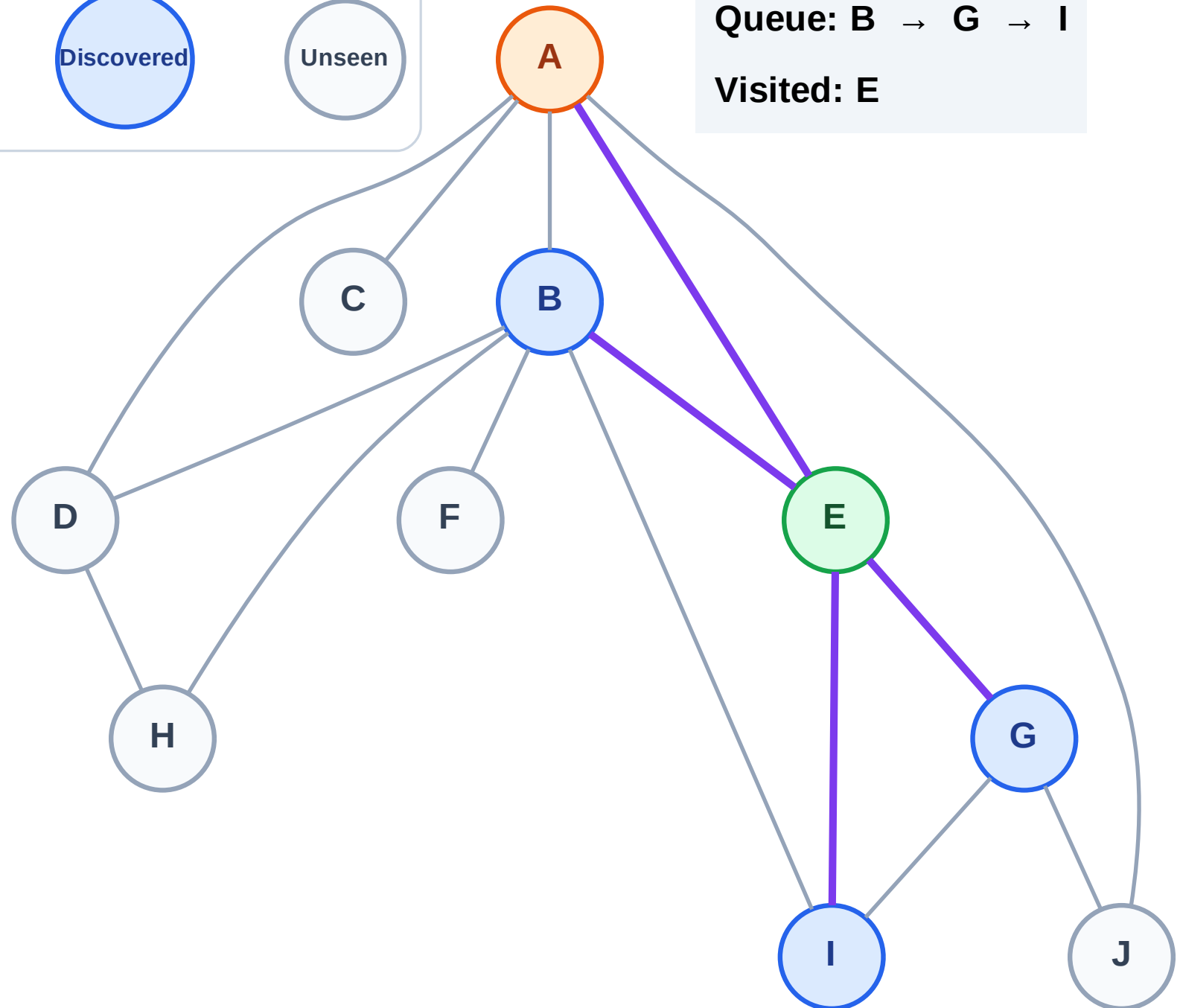
Processing

Discovered

Unseen

Queue: B → G → I

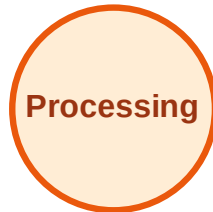
Visited: E



BFS Traversal

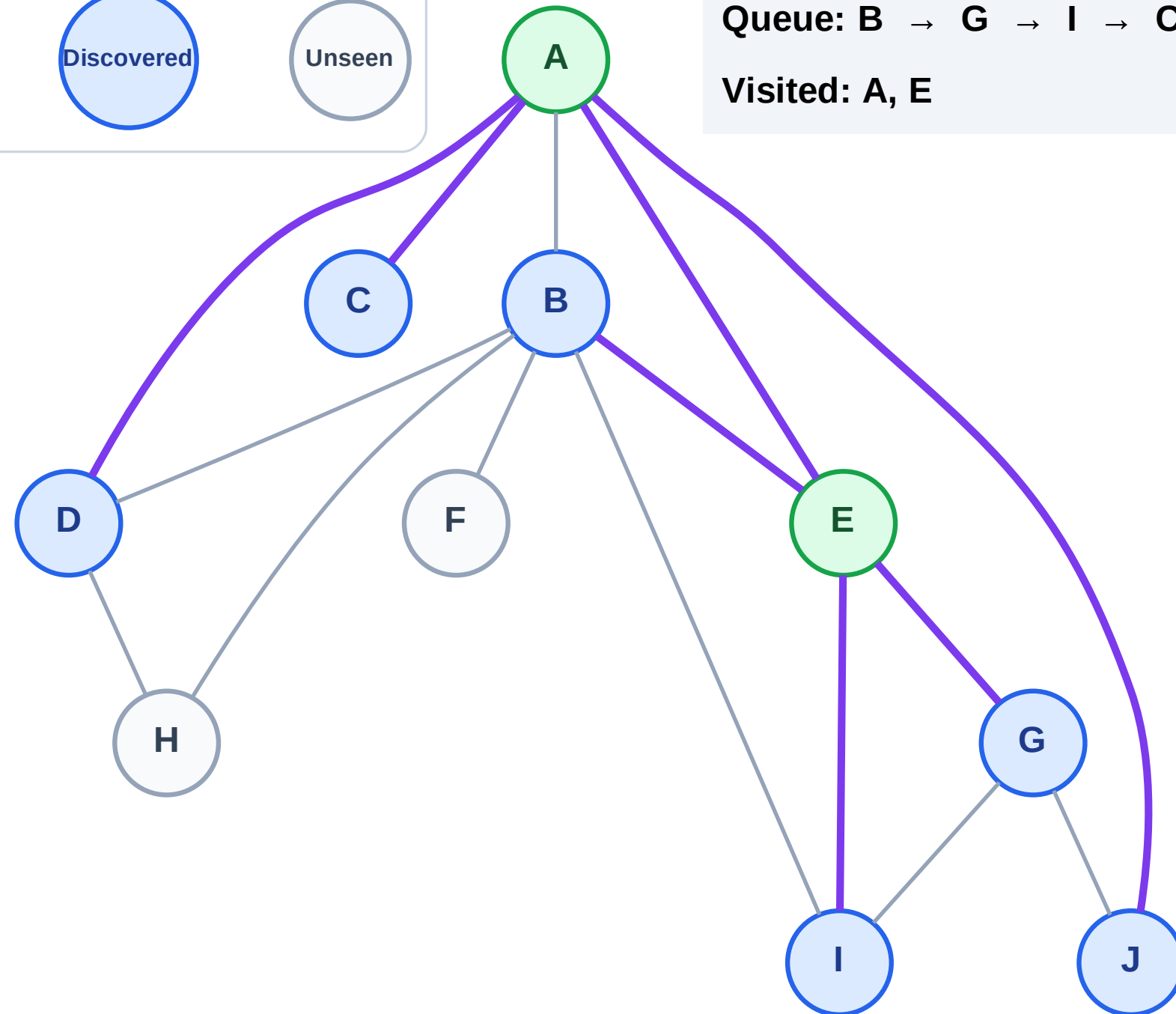
Step 5: Discover C, D, J from A

Legend



Queue: B → G → I → C → D → J

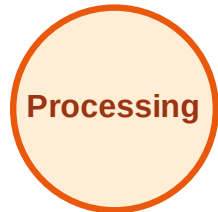
Visited: A, E



BFS Traversal

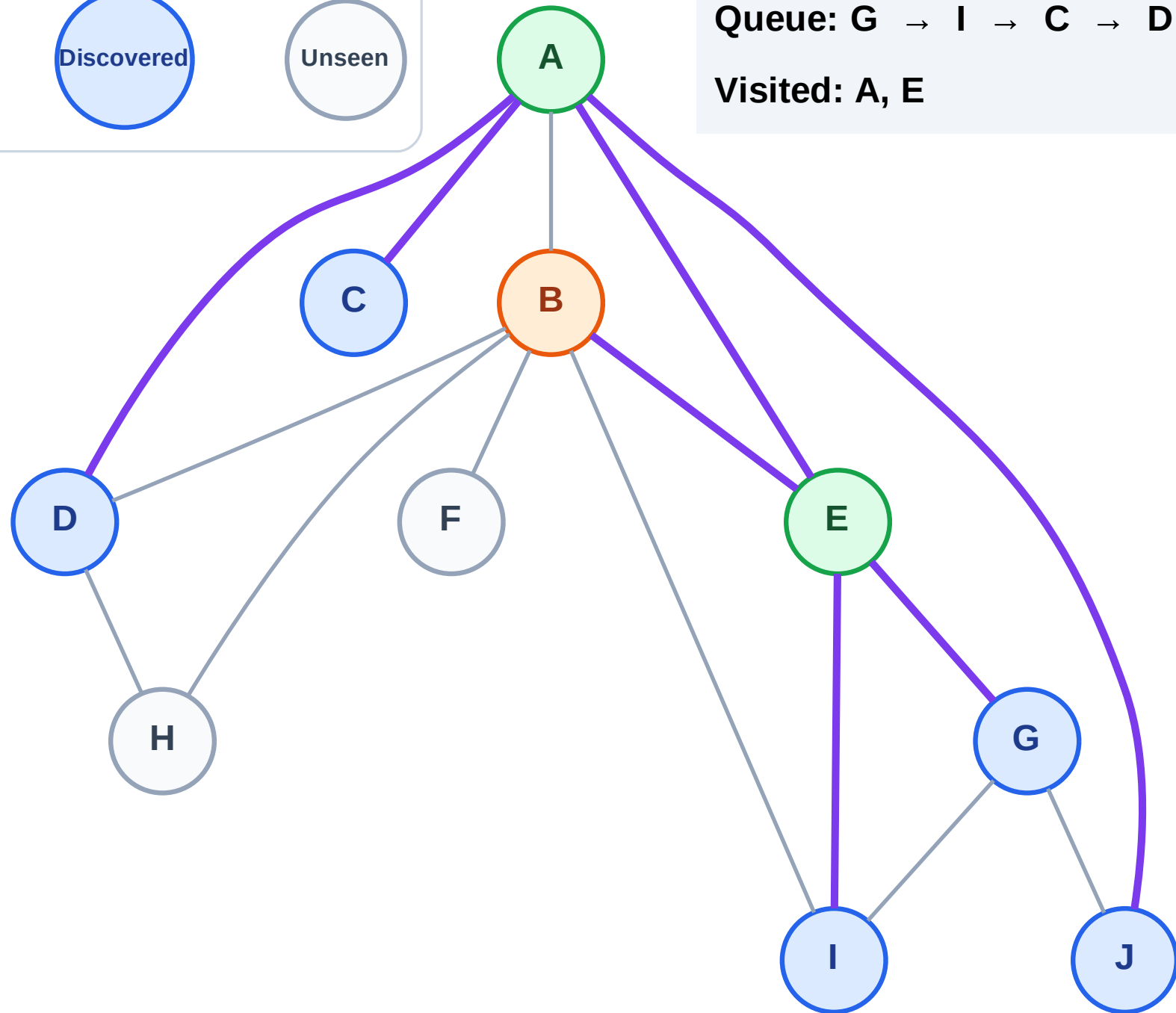
Step 6: Process node B

Legend



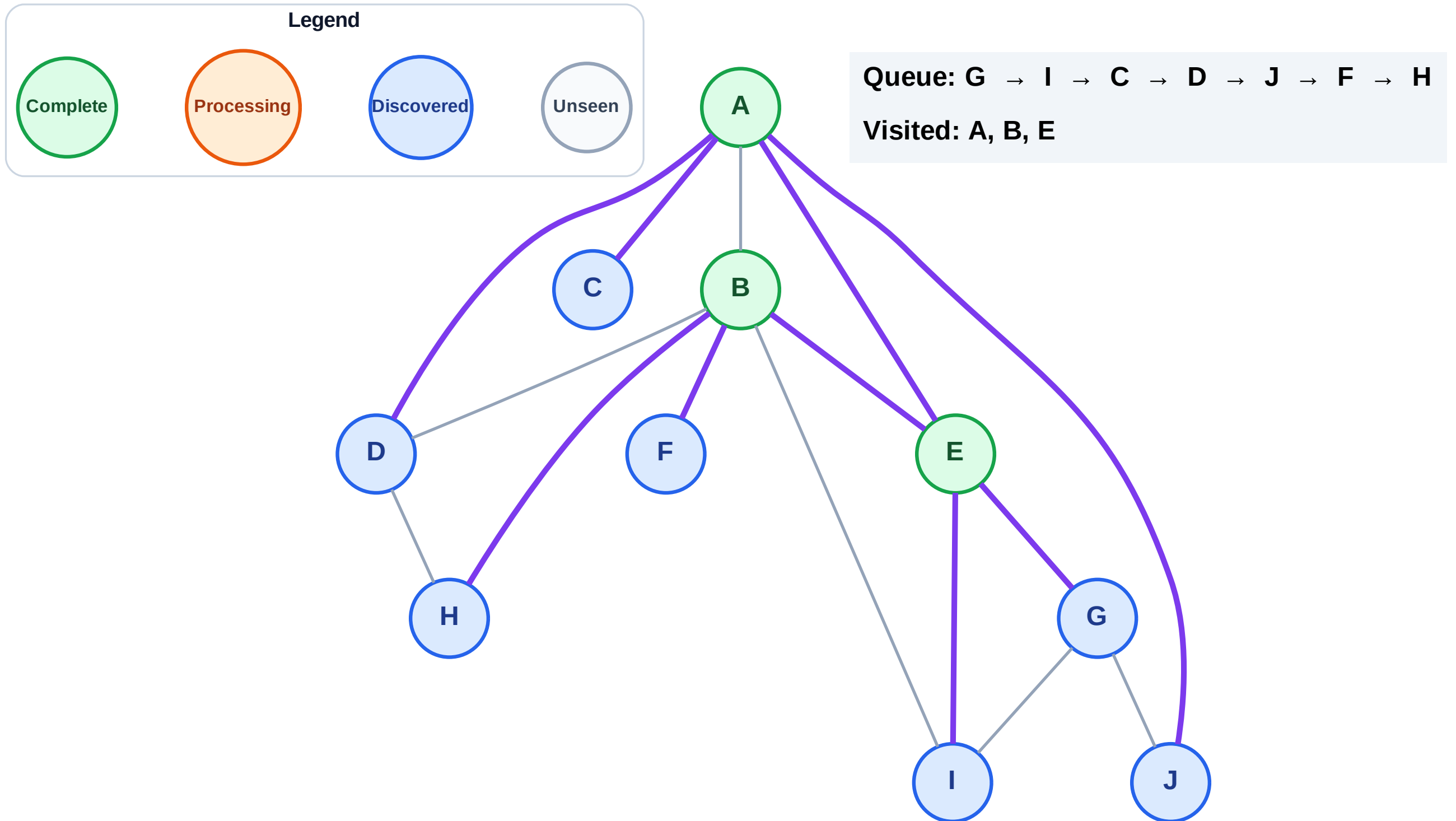
Queue: G → I → C → D → J

Visited: A, E



Step 7: Discover F, H from B

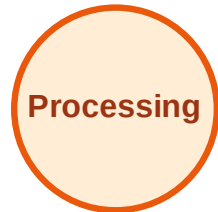
Step 7: Discover F, H from B



BFS Traversal

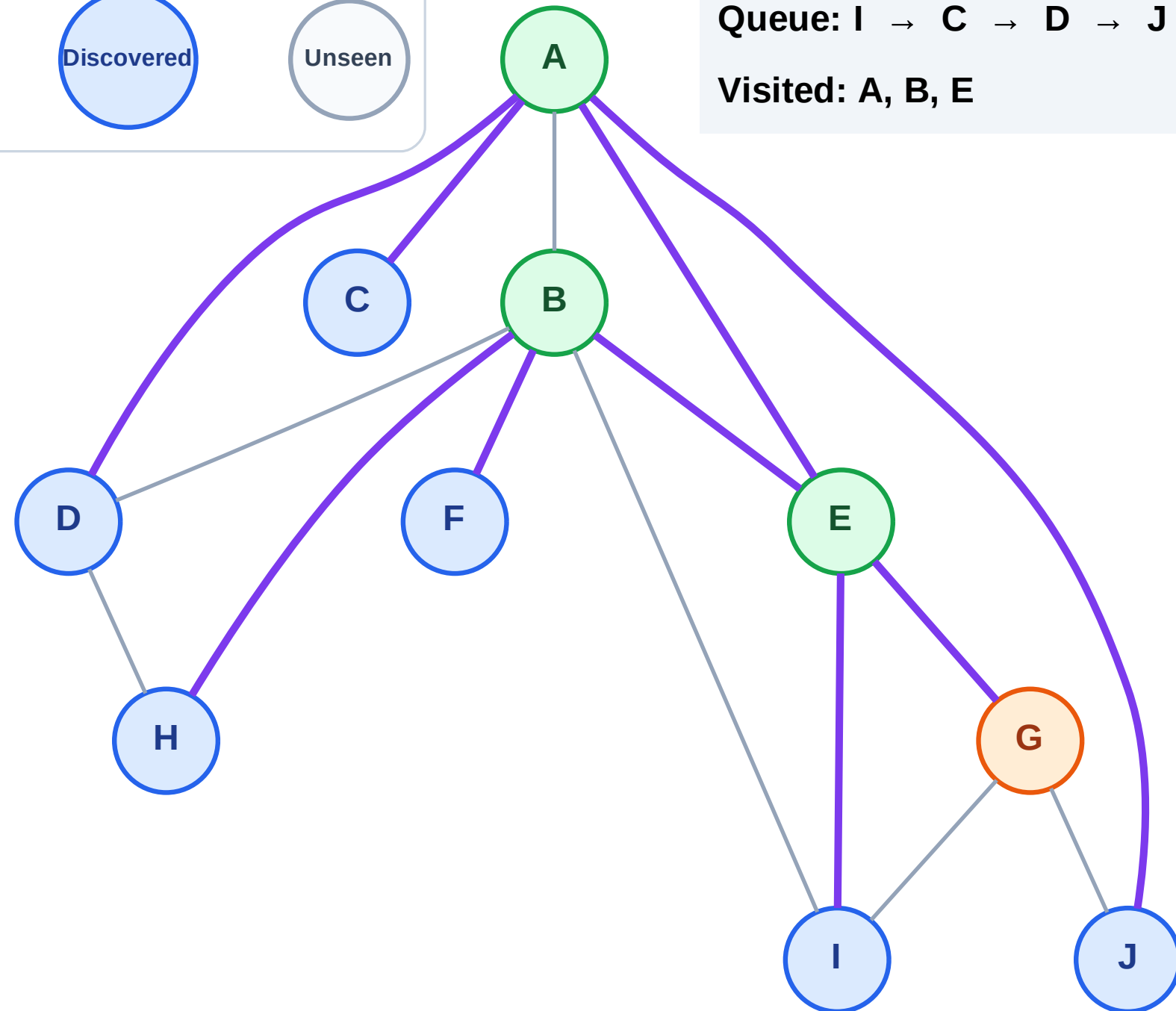
Step 8: Process node G

Legend



Queue: I → C → D → J → F → H

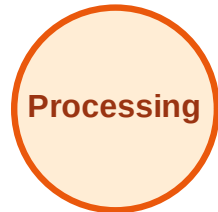
Visited: A, B, E



BFS Traversal

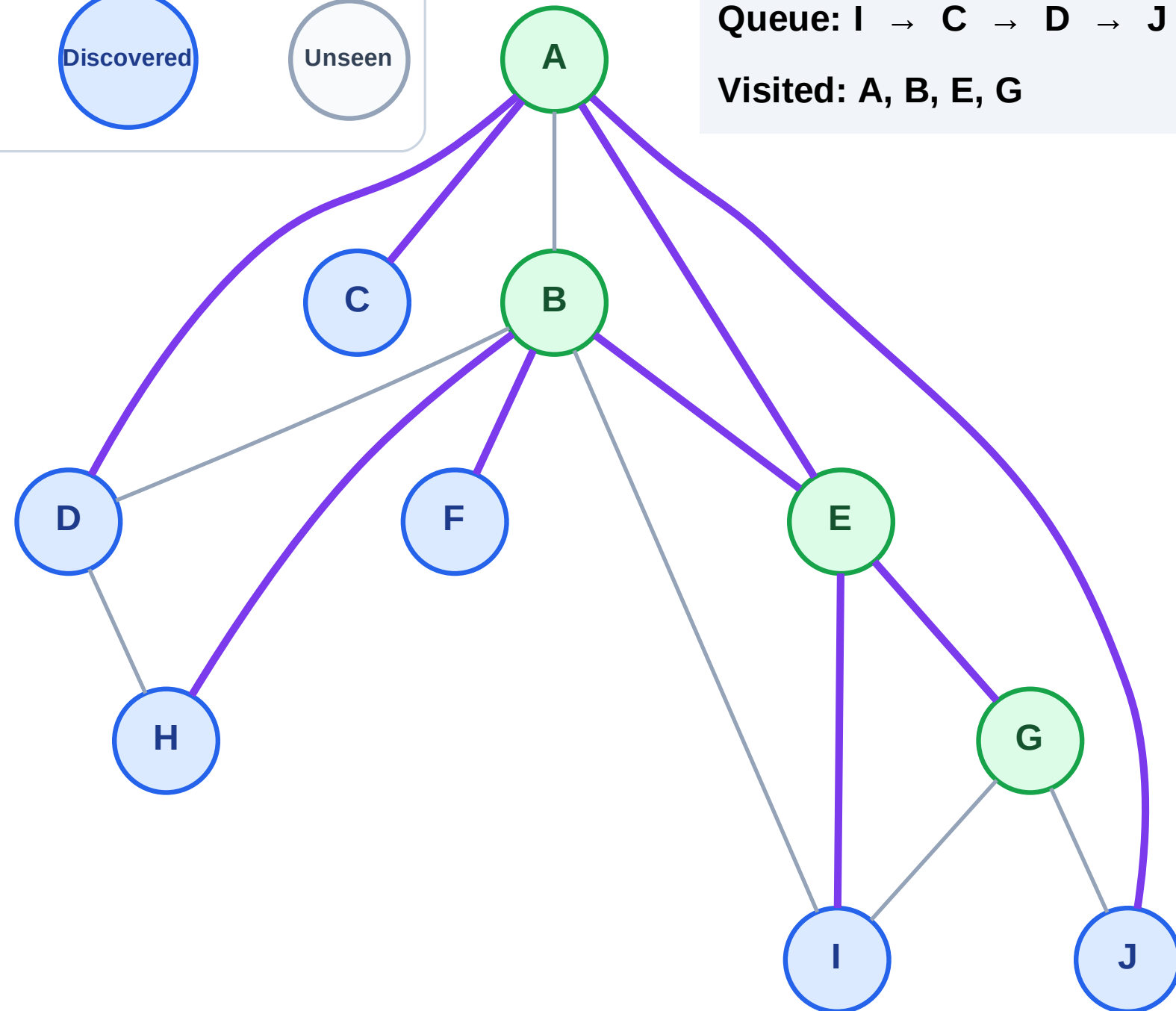
Step 9: No new neighbors from G

Legend



Queue: I → C → D → J → F → H

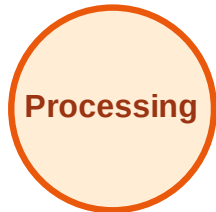
Visited: A, B, E, G



BFS Traversal

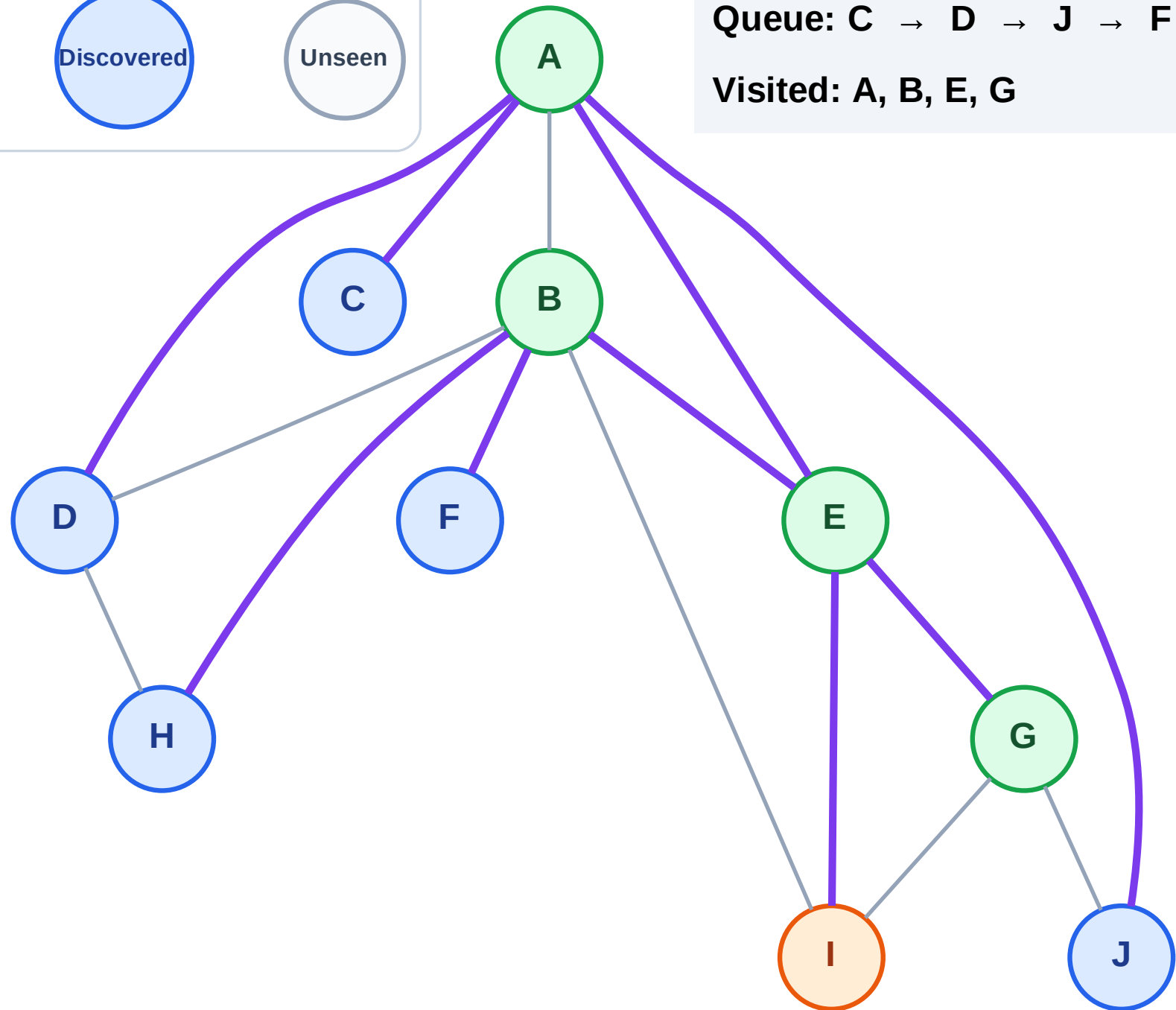
Step 10: Process node I

Legend



Queue: C → D → J → F → H

Visited: A, B, E, G



BFS Traversal

Step 11: No new neighbors from I

Legend

Complete

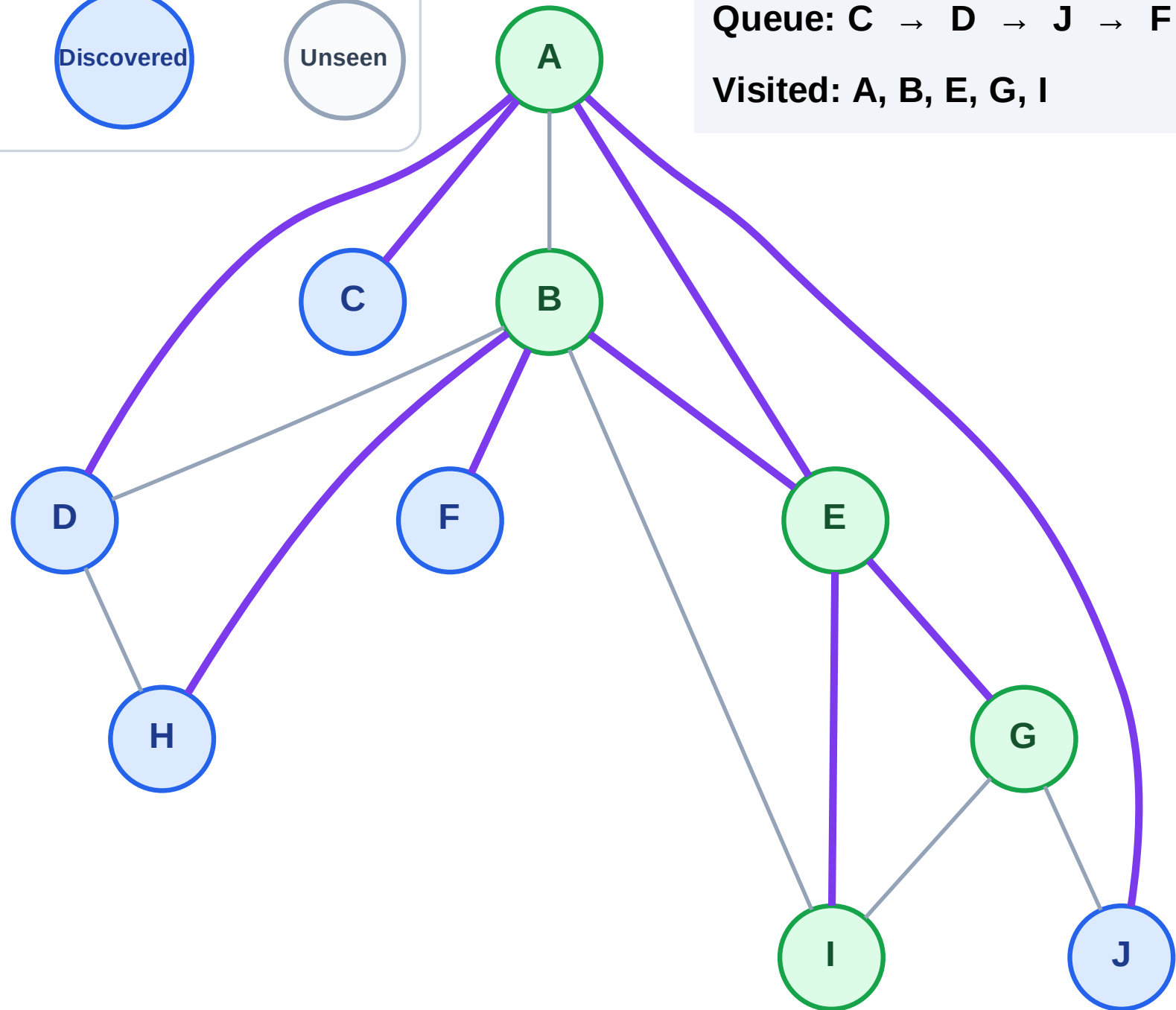
Processing

Discovered

Unseen

Queue: C → D → J → F → H

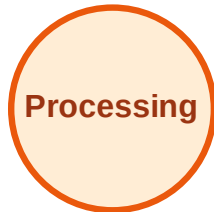
Visited: A, B, E, G, I



BFS Traversal

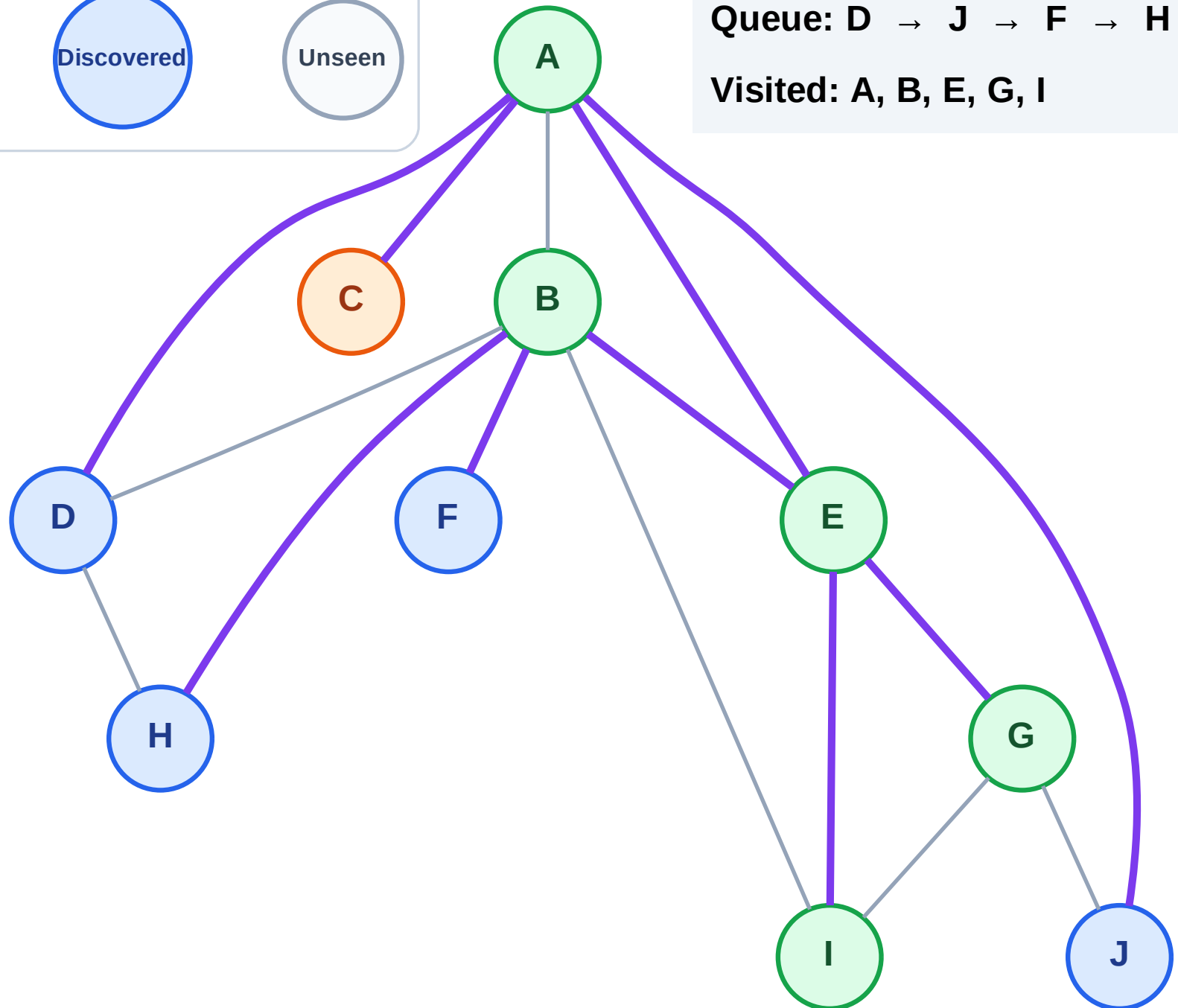
Step 12: Process node C

Legend



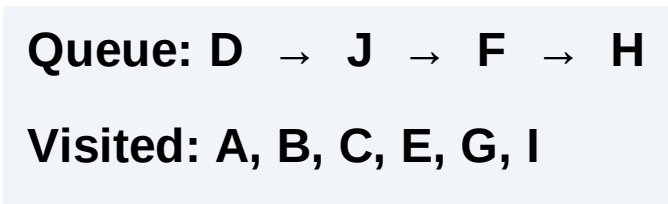
Queue: D → J → F → H

Visited: A, B, E, G, I



Step 13: No new neighbors from C

Step 13: No new neighbors from C



BFS Traversal

Step 14: Process node D

Legend

Complete

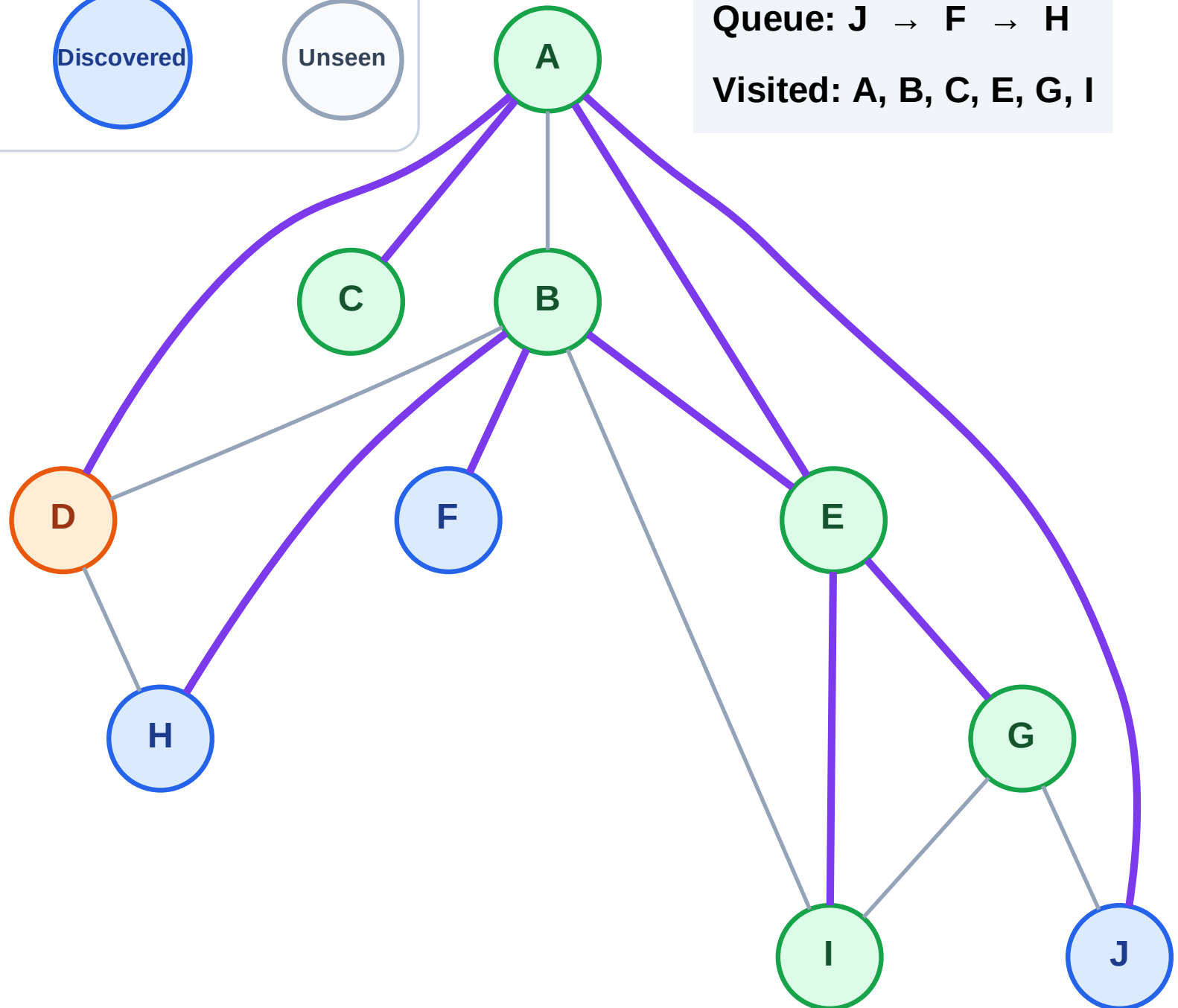
Processing

Discovered

Unseen

Queue: J → F → H

Visited: A, B, C, E, G, I



BFS Traversal

Step 15: No new neighbors from D

Legend

Complete

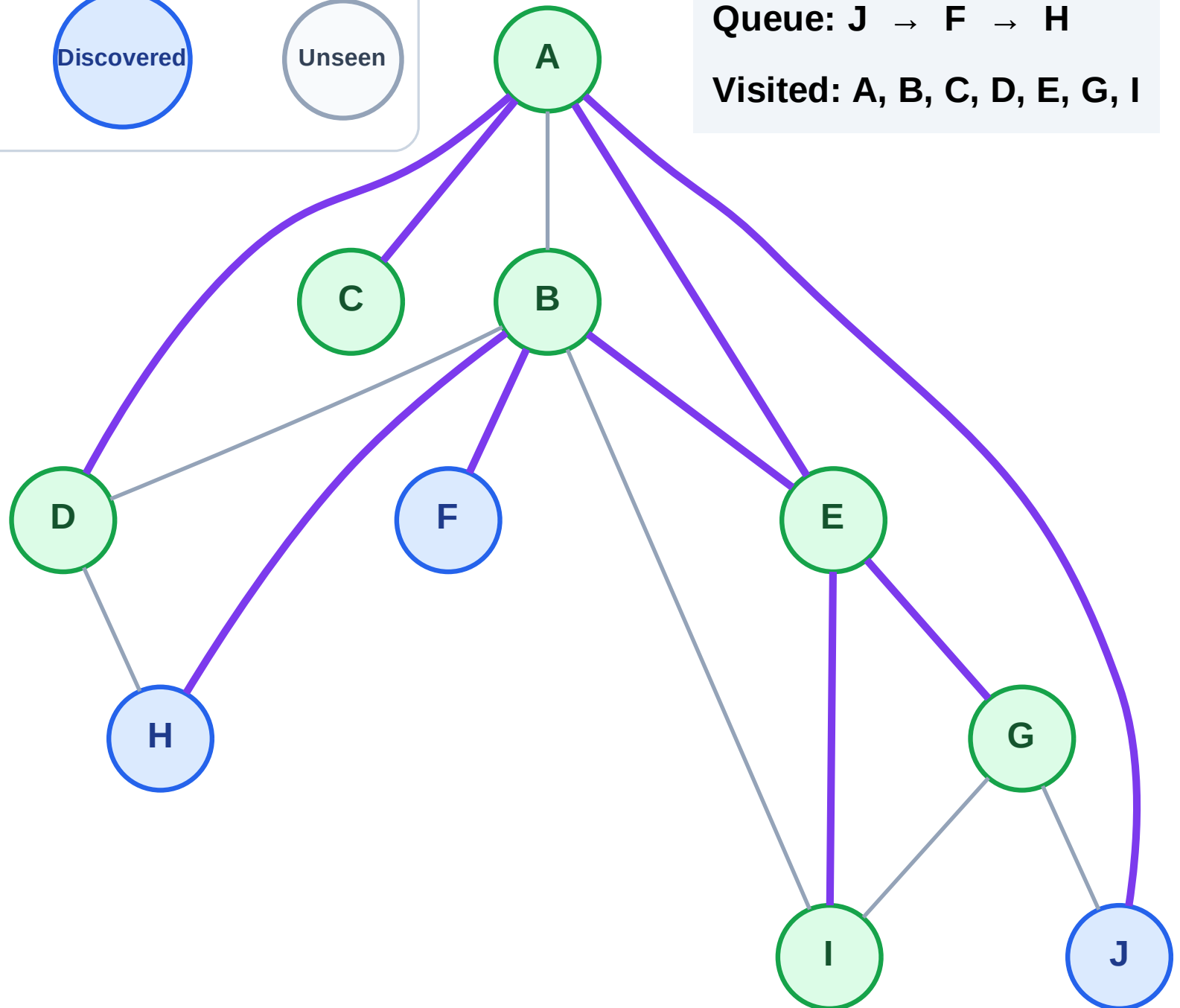
Processing

Discovered

Unseen

Queue: J → F → H

Visited: A, B, C, D, E, G, I



BFS Traversal

Step 16: Process node J

Legend

Complete

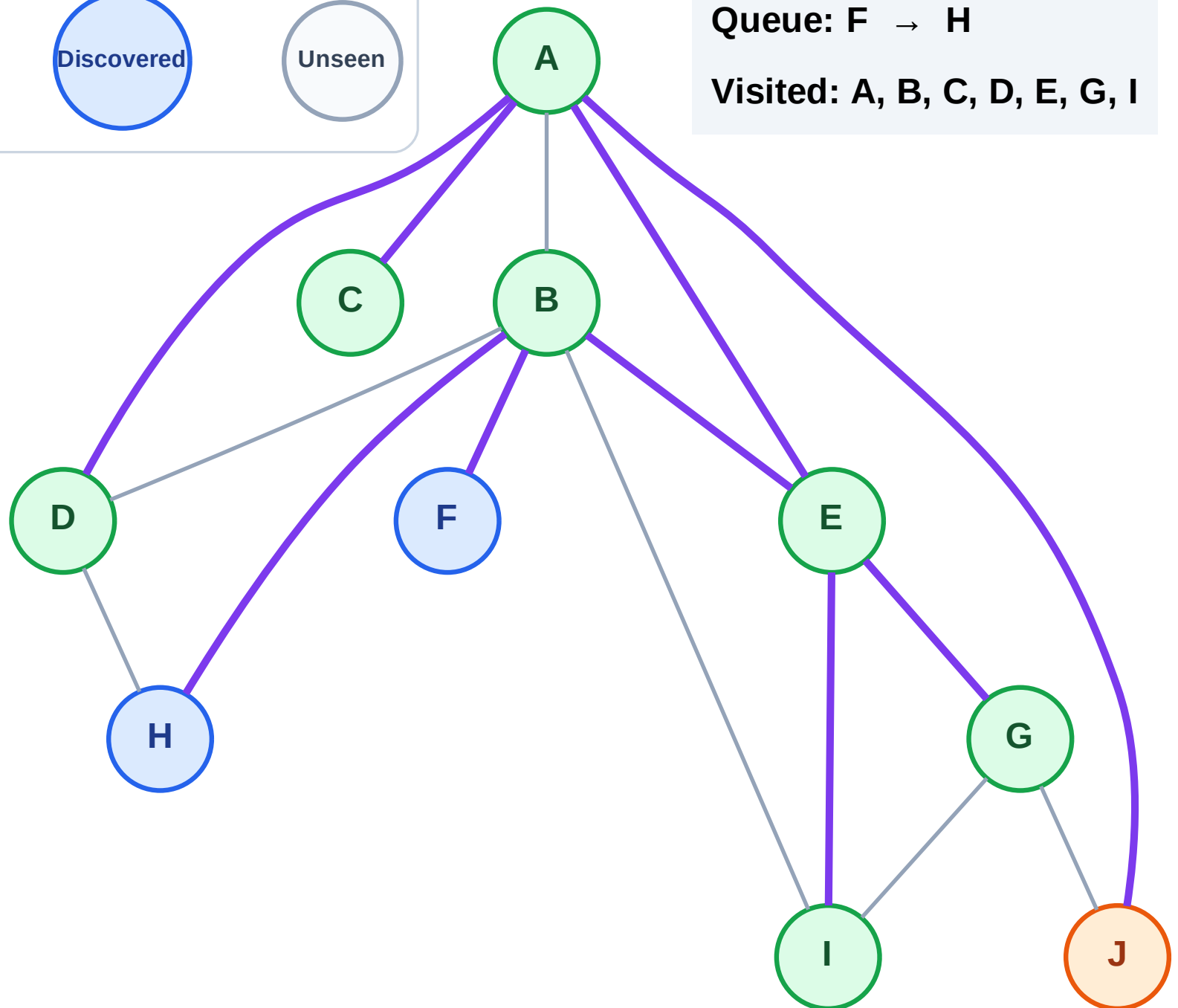
Processing

Discovered

Unseen

Queue: F → H

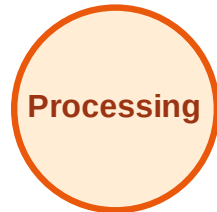
Visited: A, B, C, D, E, G, I



BFS Traversal

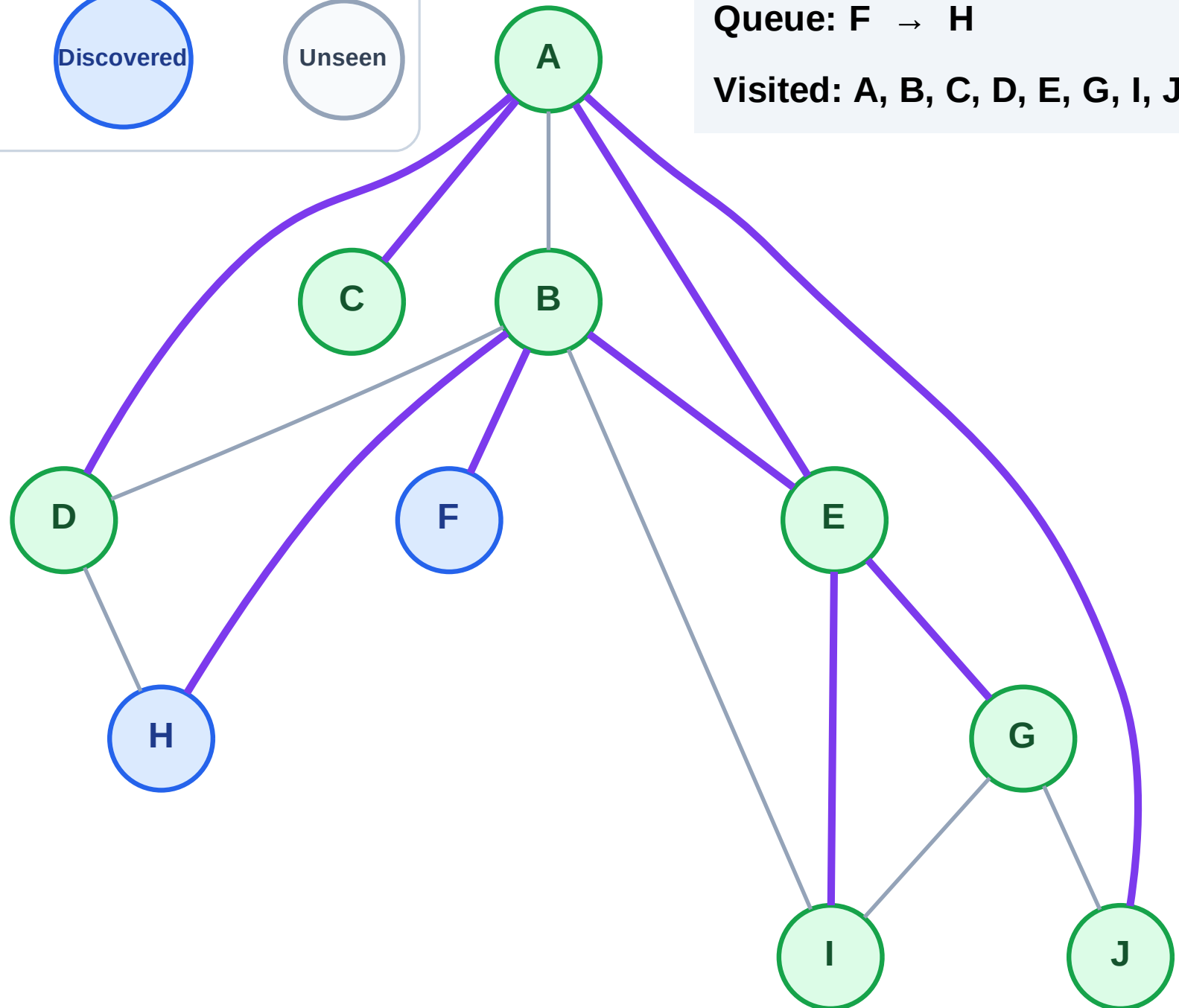
Step 17: No new neighbors from J

Legend



Queue: F → H

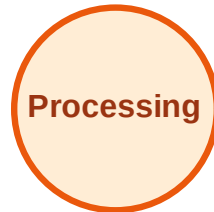
Visited: A, B, C, D, E, G, I, J



BFS Traversal

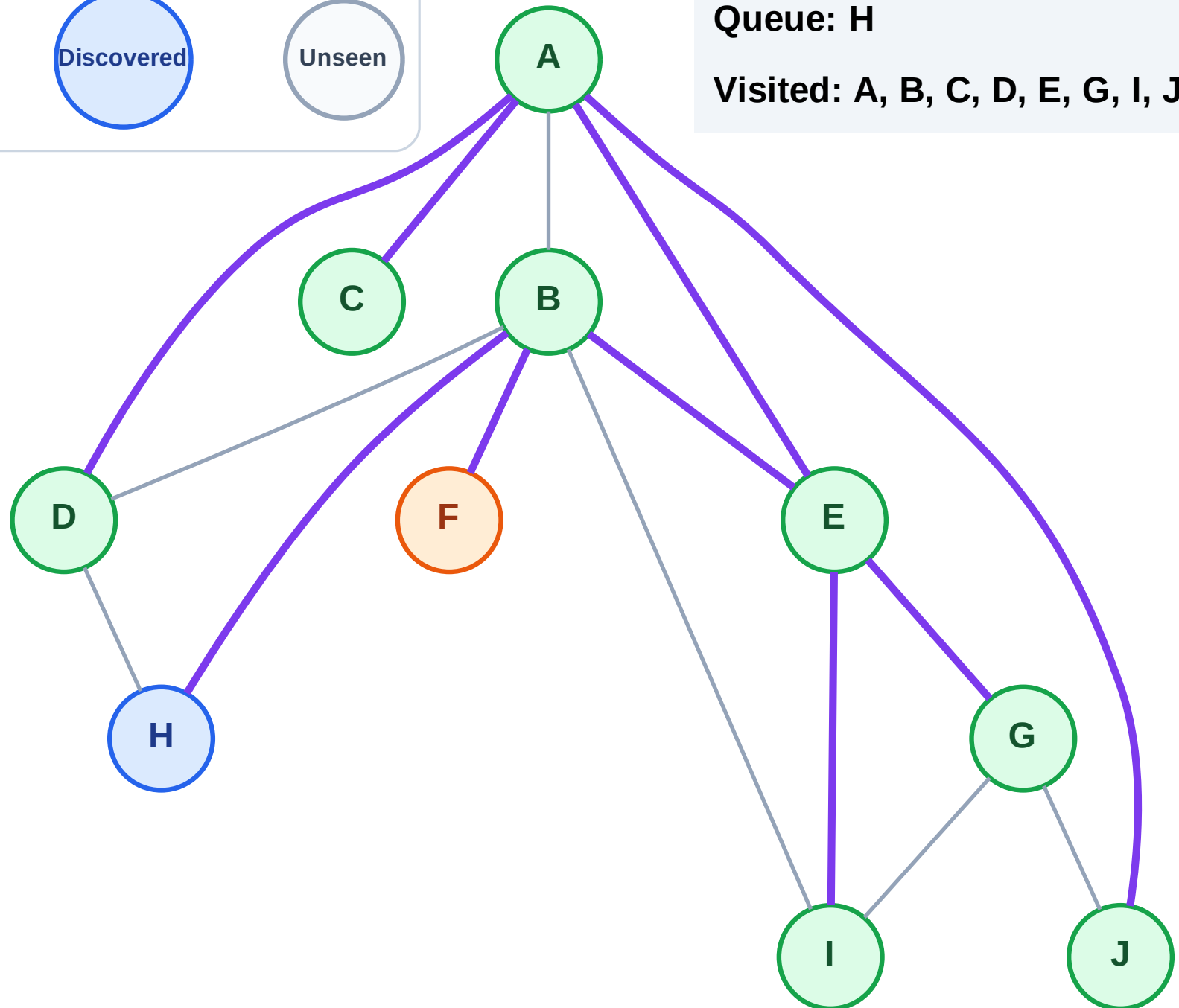
Step 18: Process node F

Legend



Queue: H

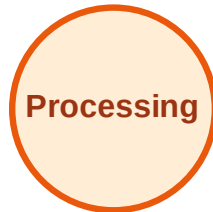
Visited: A, B, C, D, E, G, I, J



BFS Traversal

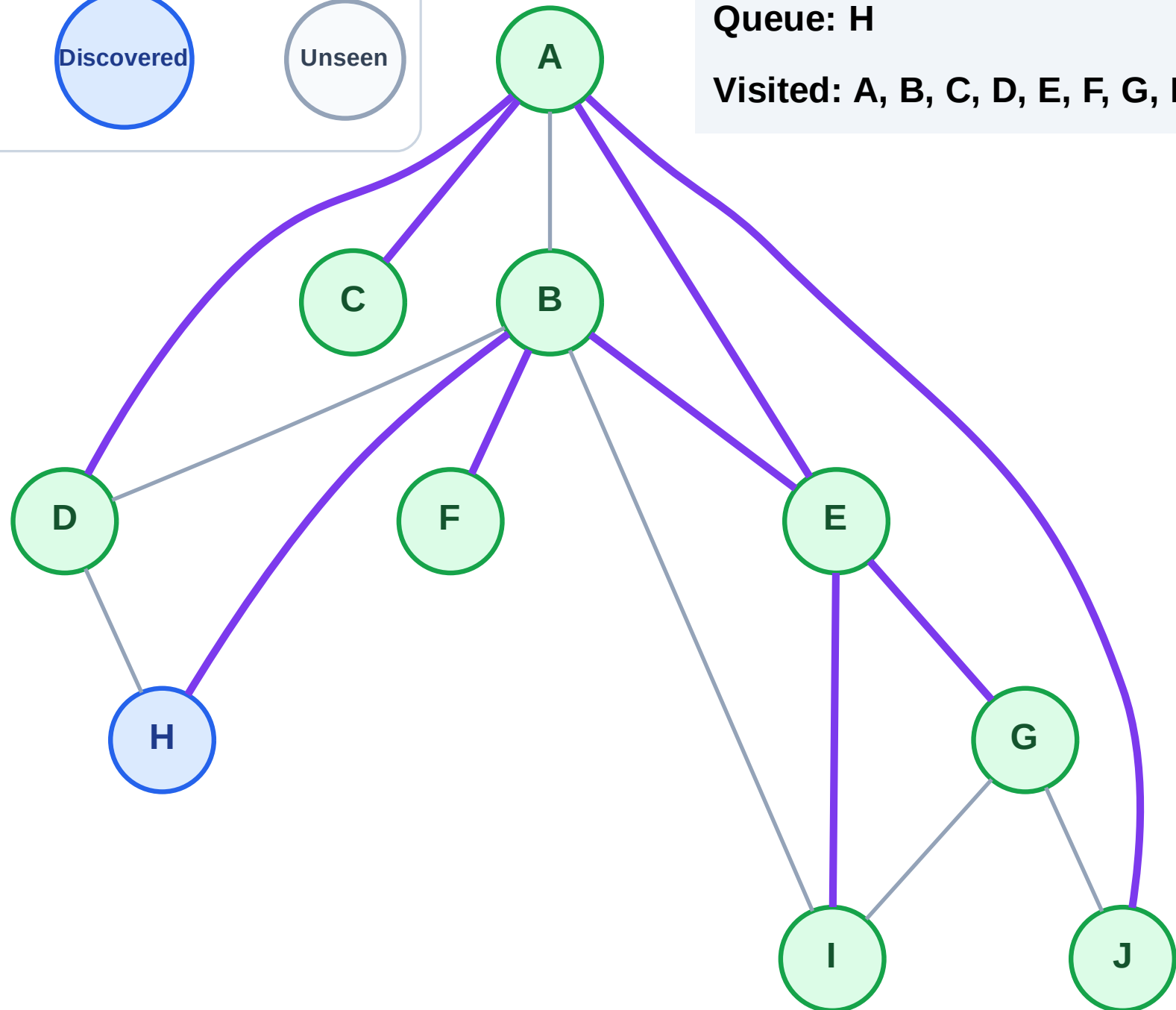
Step 19: No new neighbors from F

Legend



Queue: H

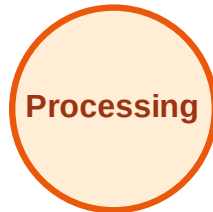
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

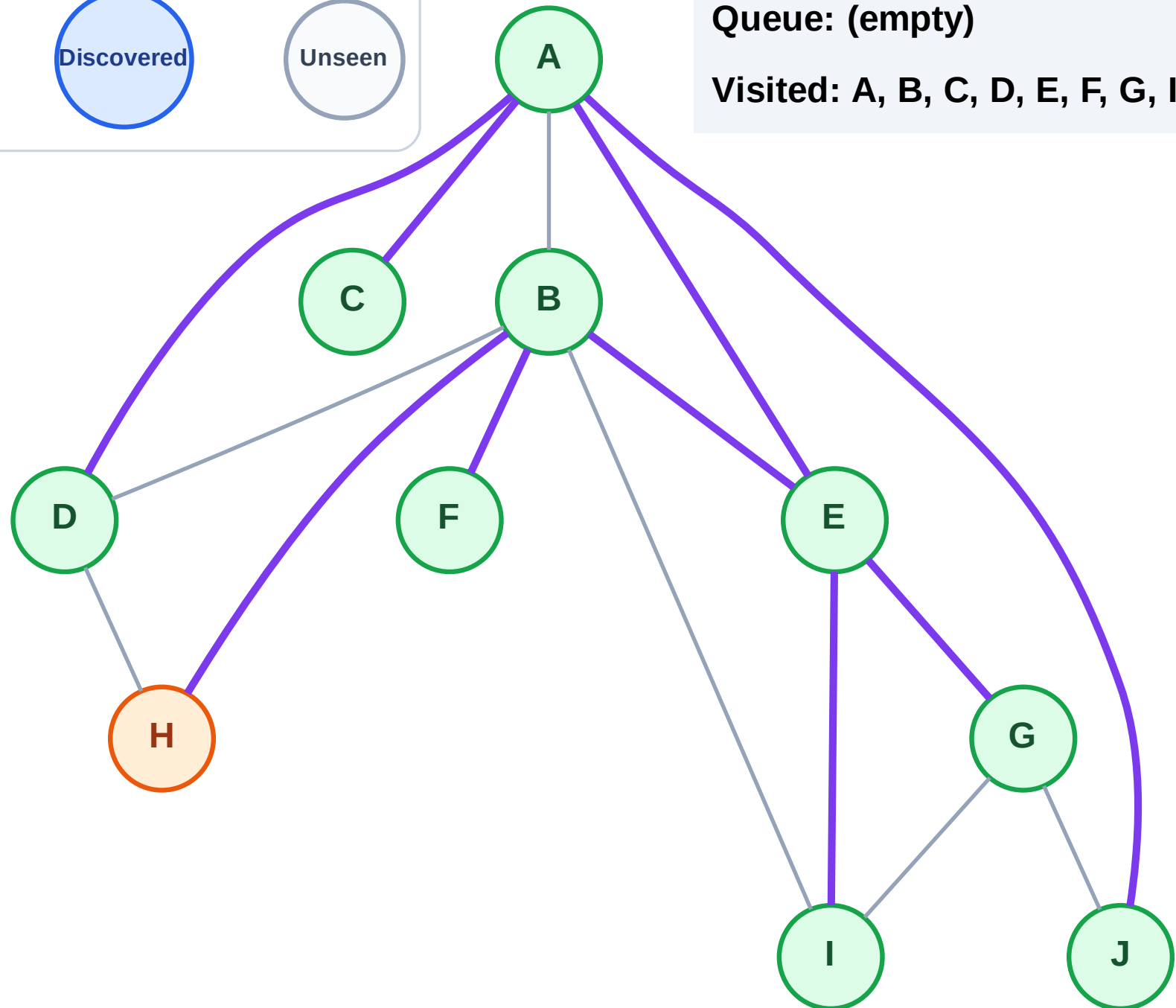
Step 20: Process node H

Legend



Queue: (empty)

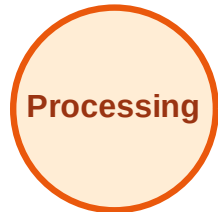
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

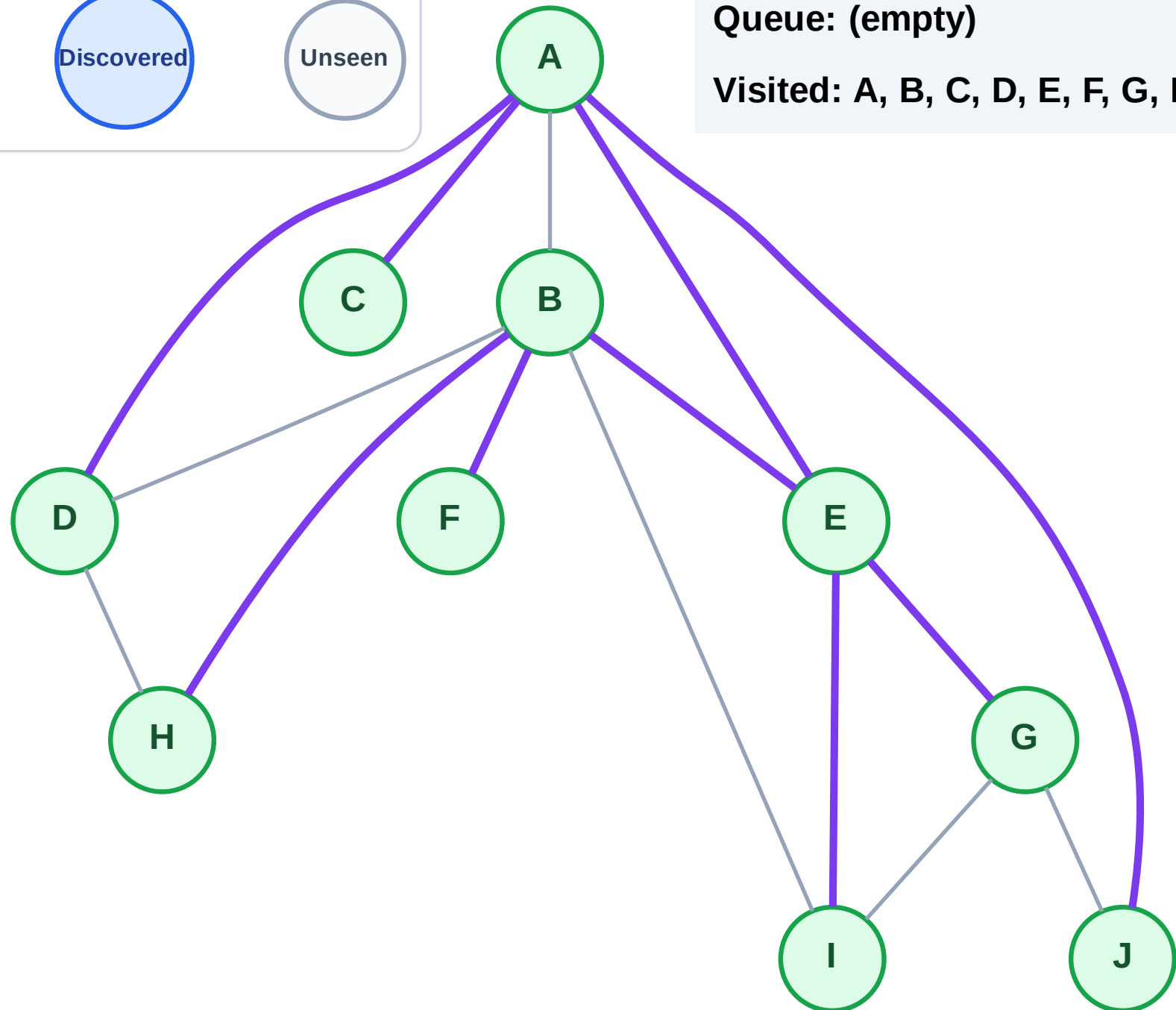
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

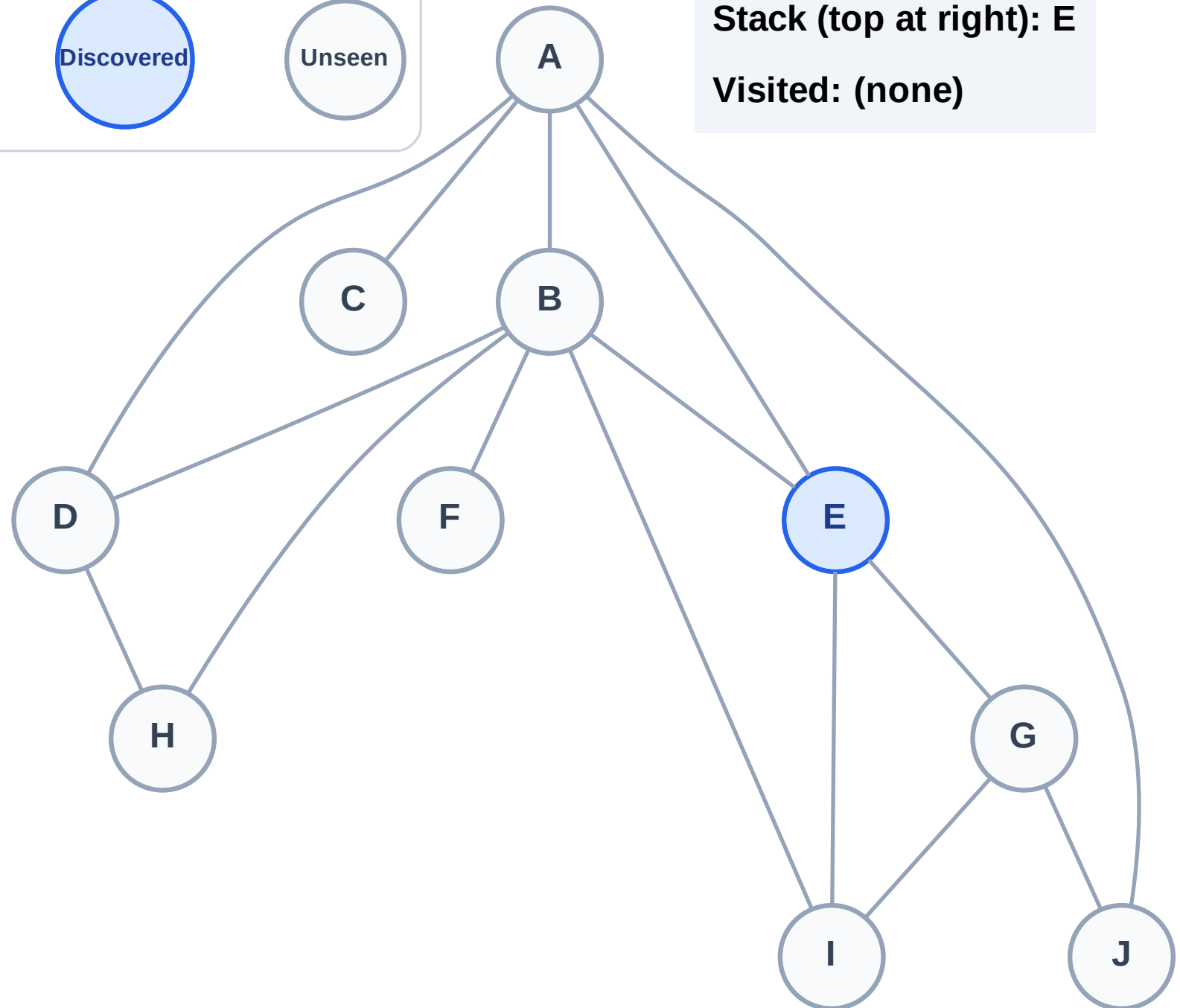
Processing

Discovered

Unseen

Stack (top at right): E

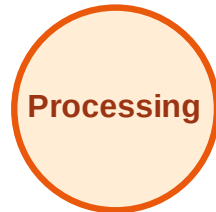
Visited: (none)



DFS Traversal

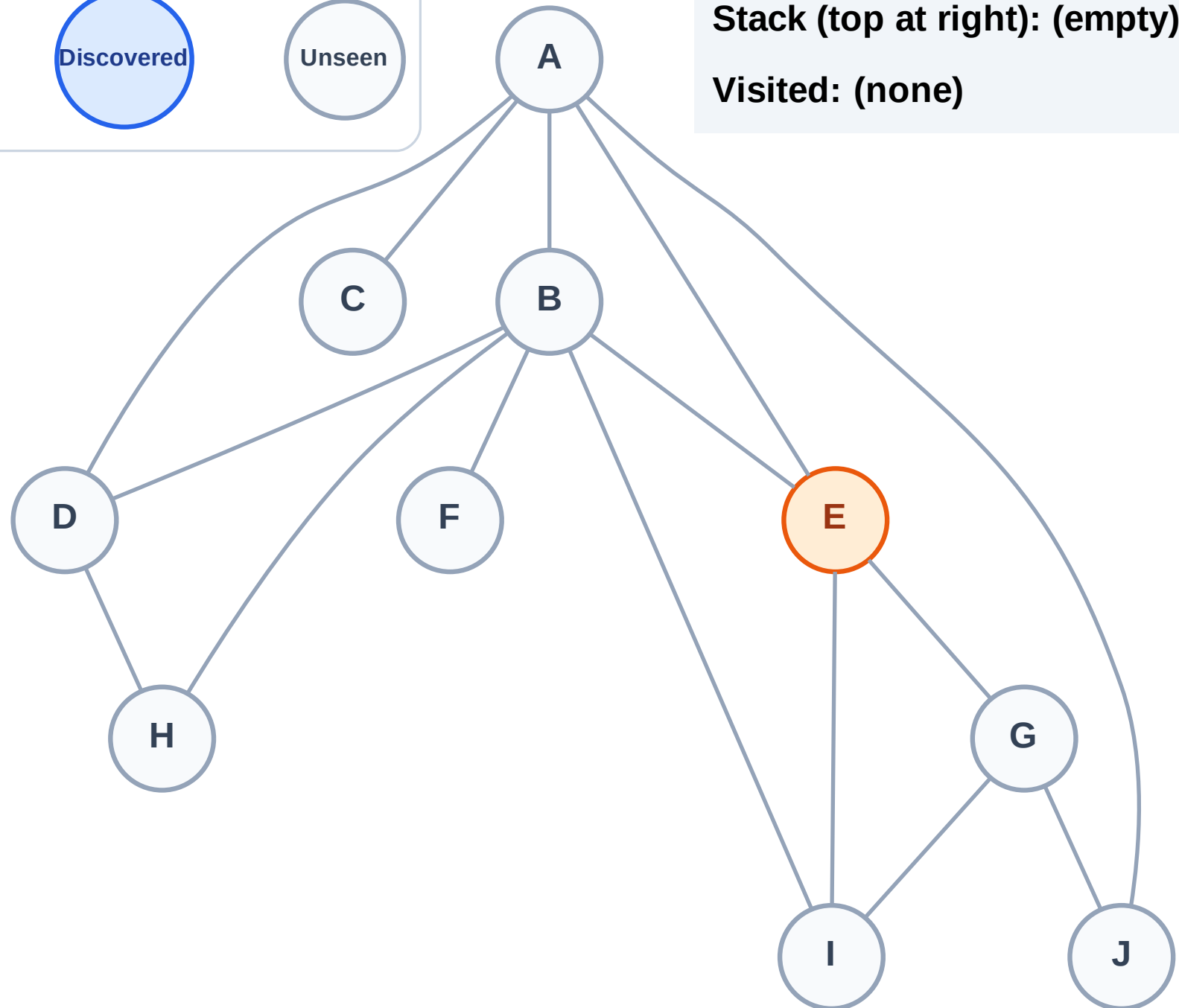
Step 2: Process node E

Legend



Stack (top at right): (empty)

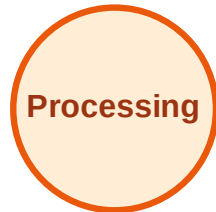
Visited: (none)



DFS Traversal

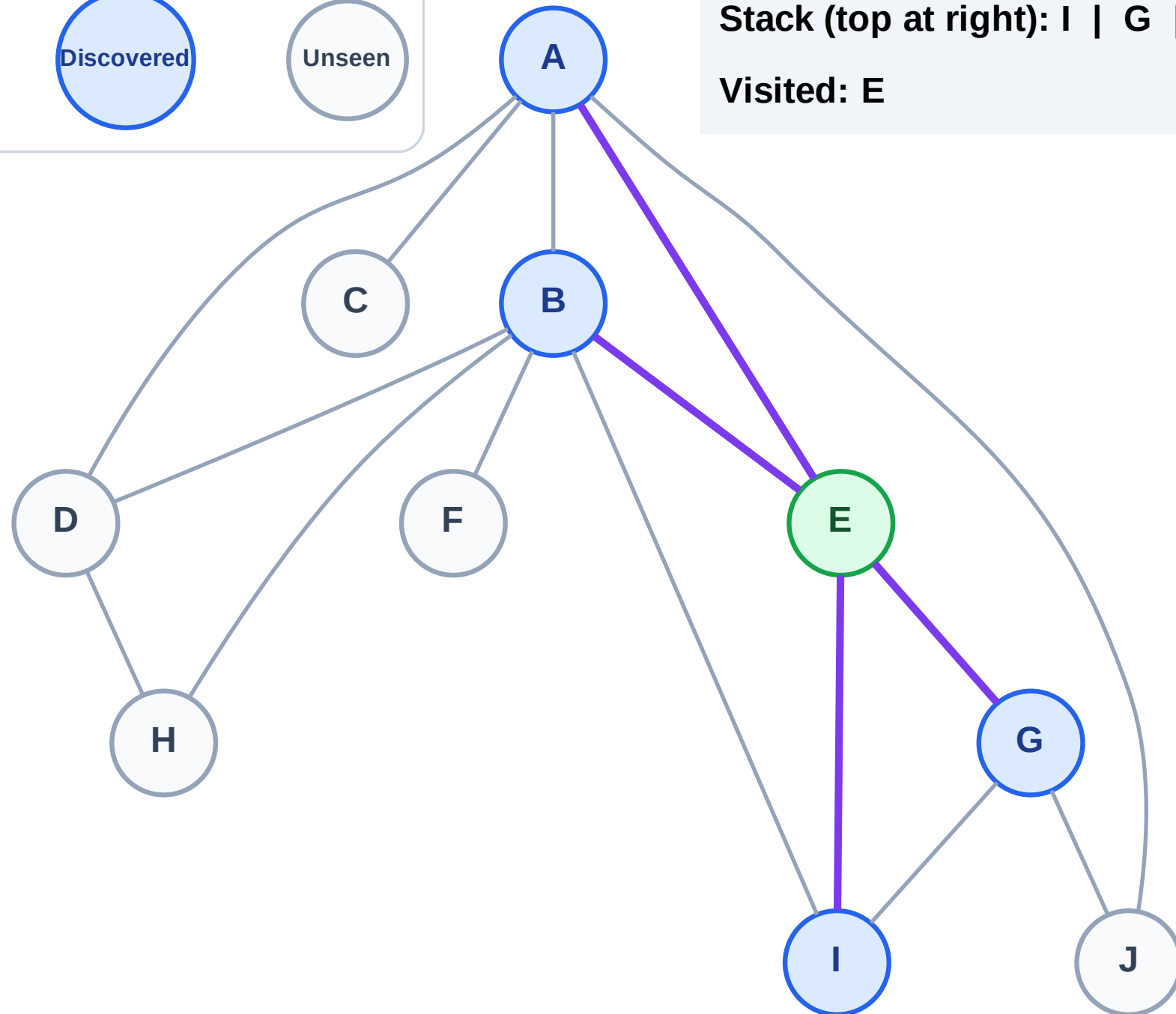
Step 3: Discover I, G, B, A from E

Legend



Stack (top at right): I | G | B | A

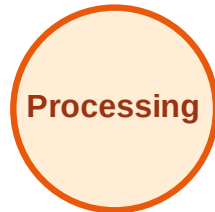
Visited: E



DFS Traversal

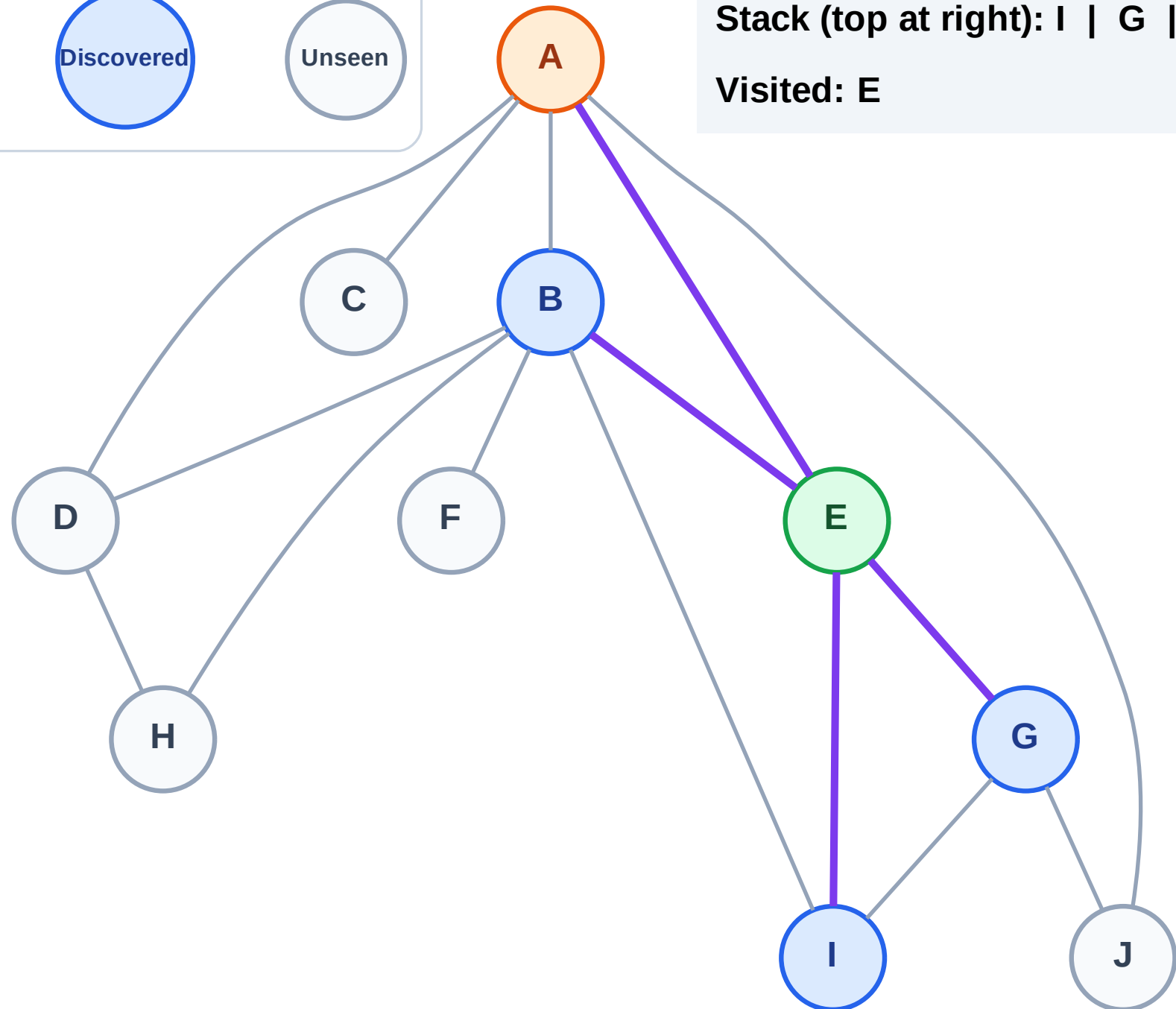
Step 4: Process node A

Legend



Stack (top at right): I | G | B

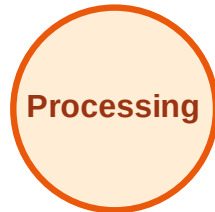
Visited: E



DFS Traversal

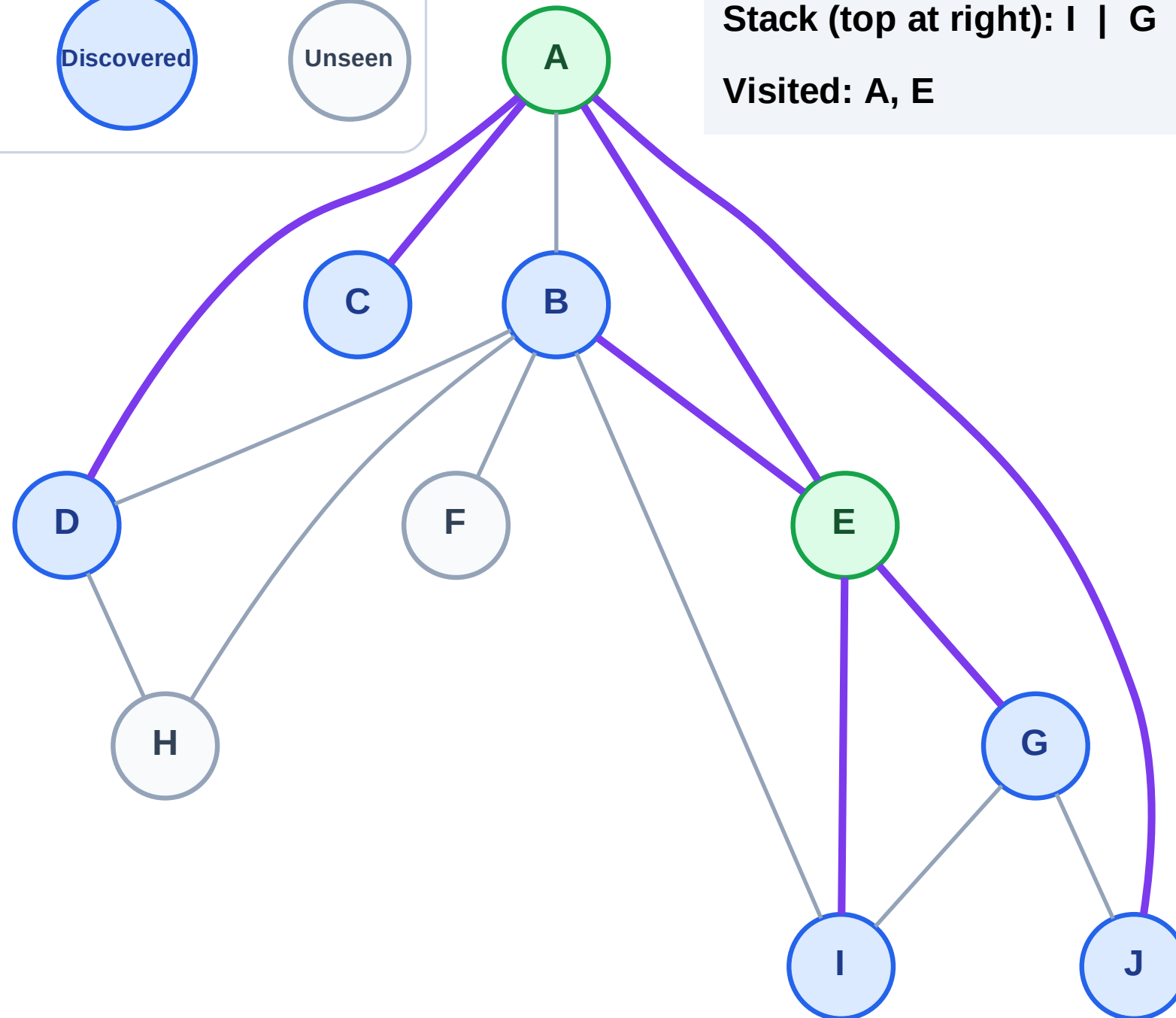
Step 5: Discover J, D, C from A

Legend



Stack (top at right): I | G | B | J | D | C

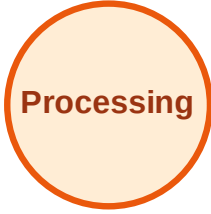
Visited: A, E



DFS Traversal

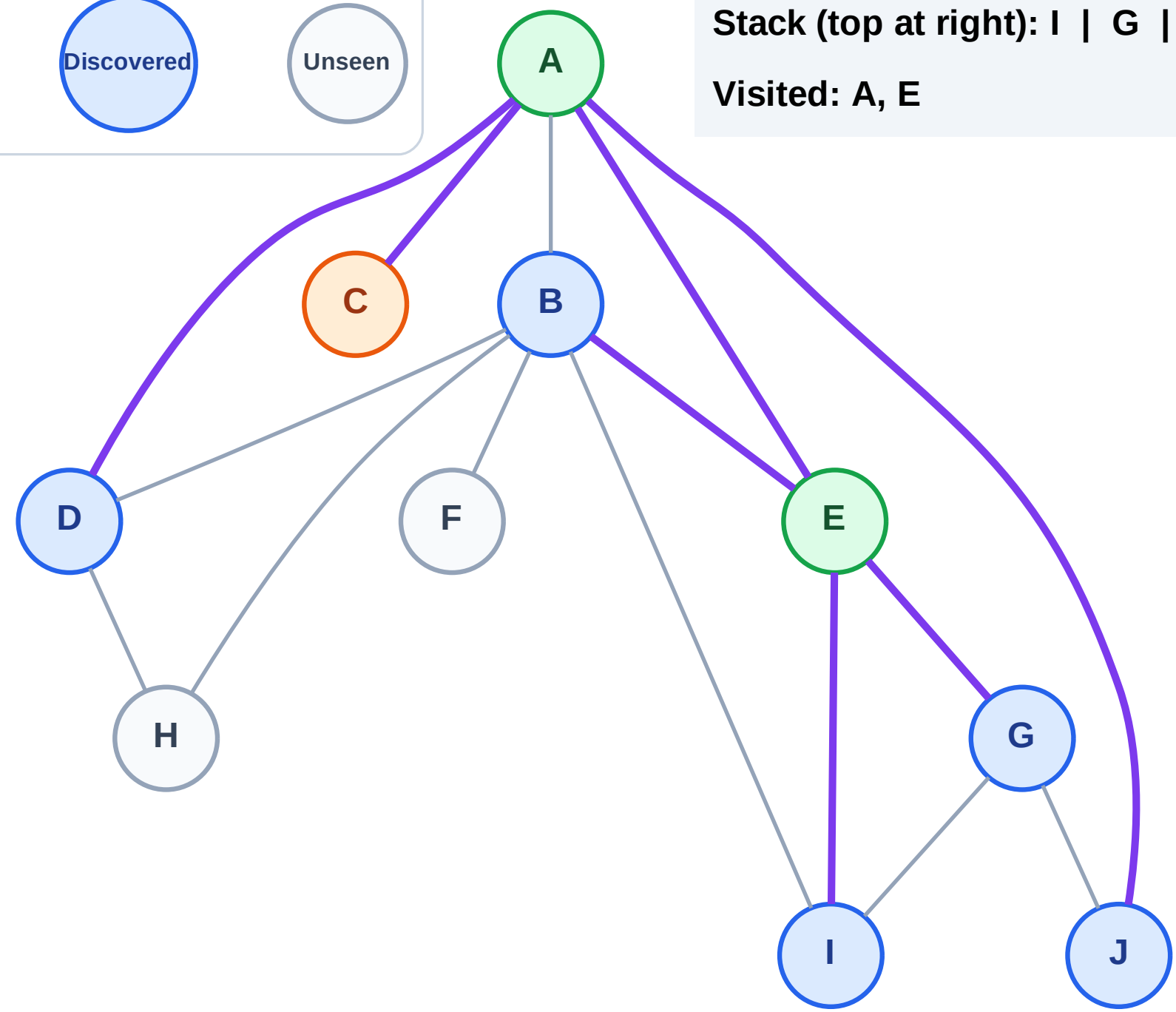
Step 6: Process node C

Legend



Stack (top at right): I | G | B | J | D

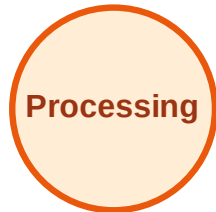
Visited: A, E



DFS Traversal

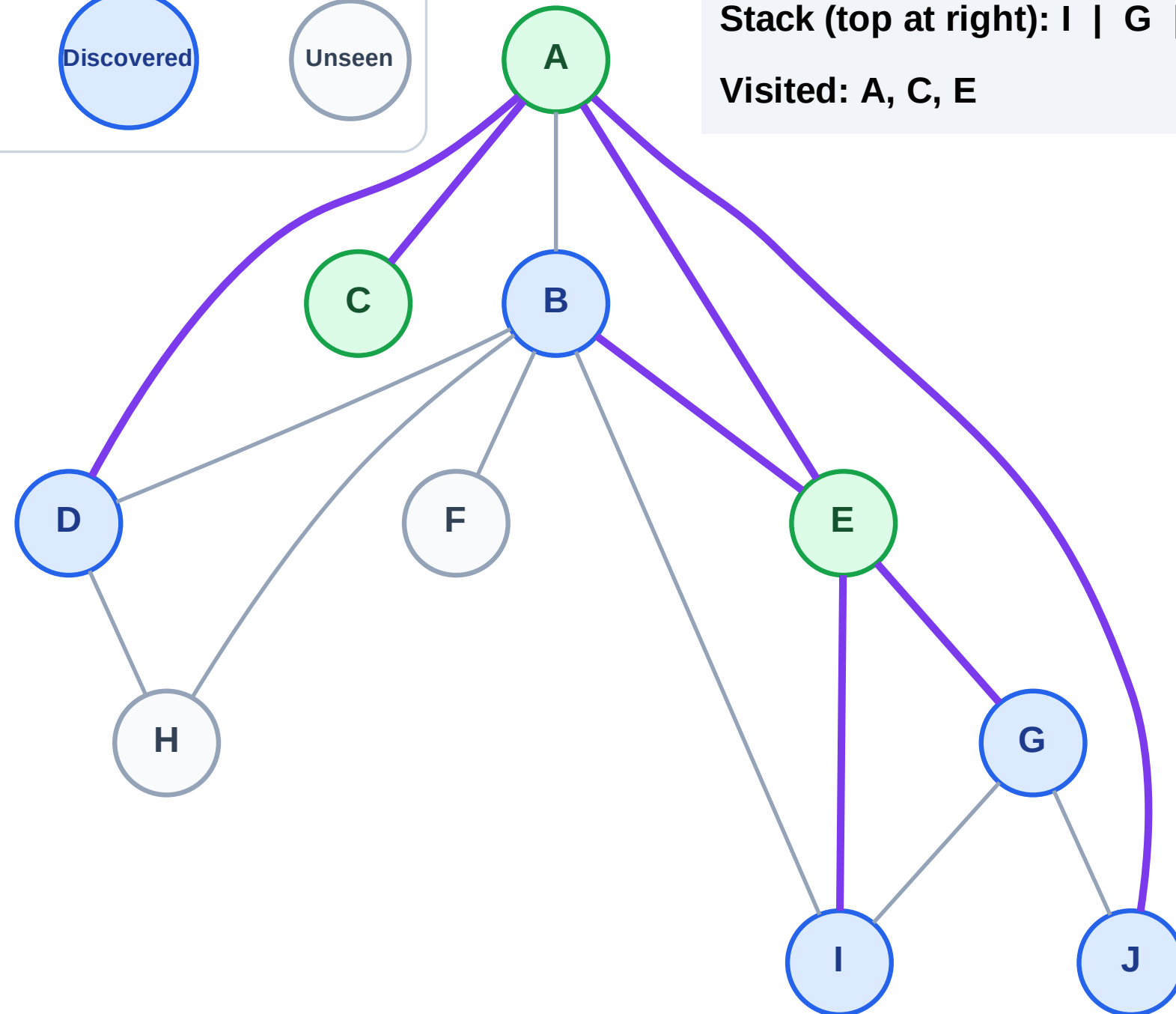
Step 7: No new neighbors from C

Legend



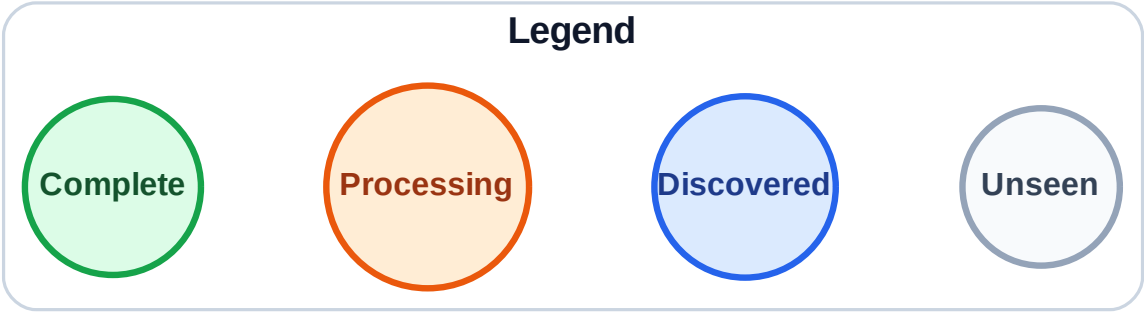
Stack (top at right): I | G | B | J | D

Visited: A, C, E

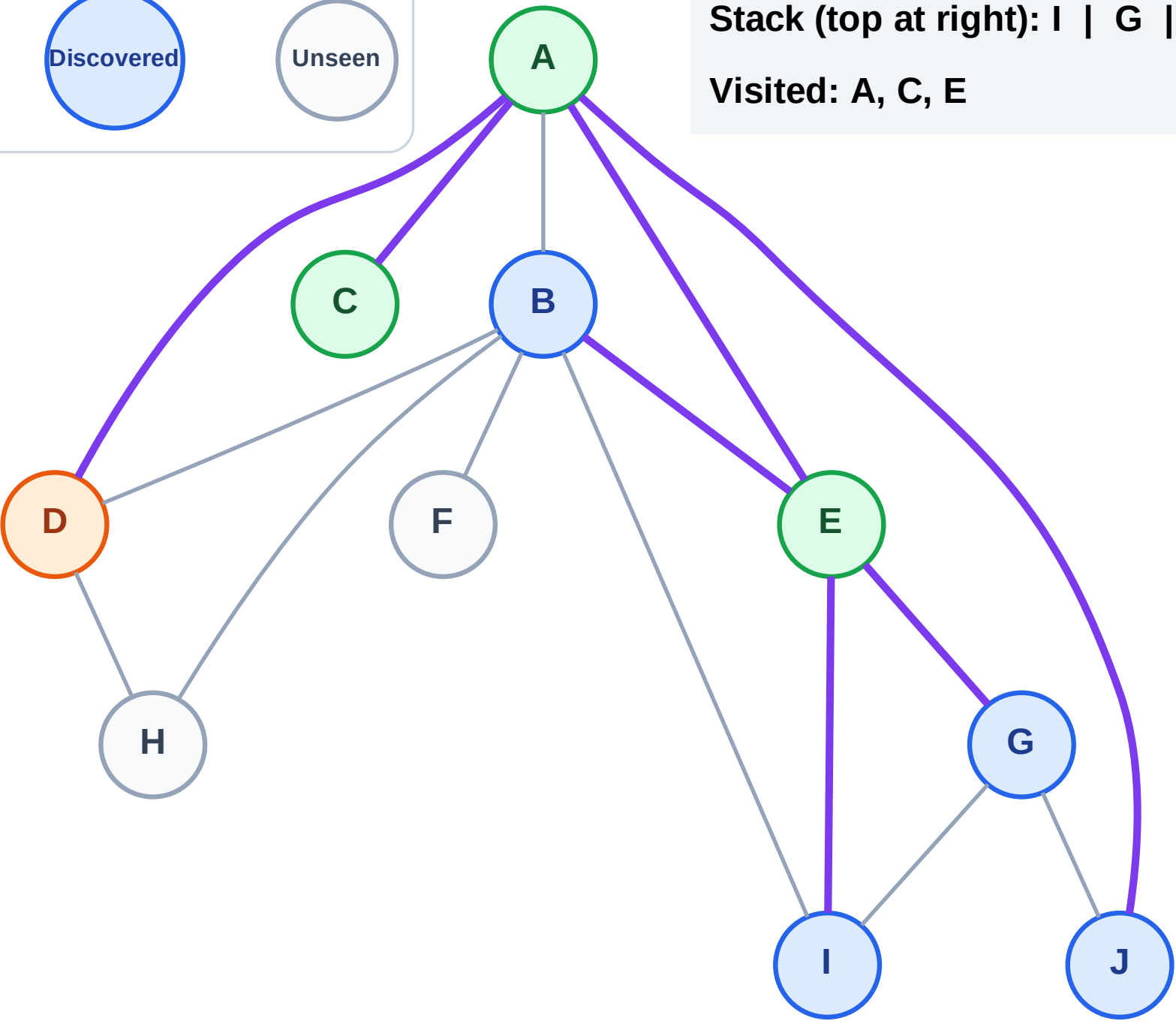


DFS Traversal

Step 8: Process node D



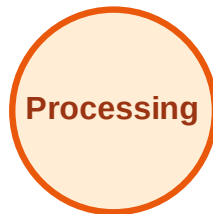
Stack (top at right): I | G | B | J
Visited: A, C, E



DFS Traversal

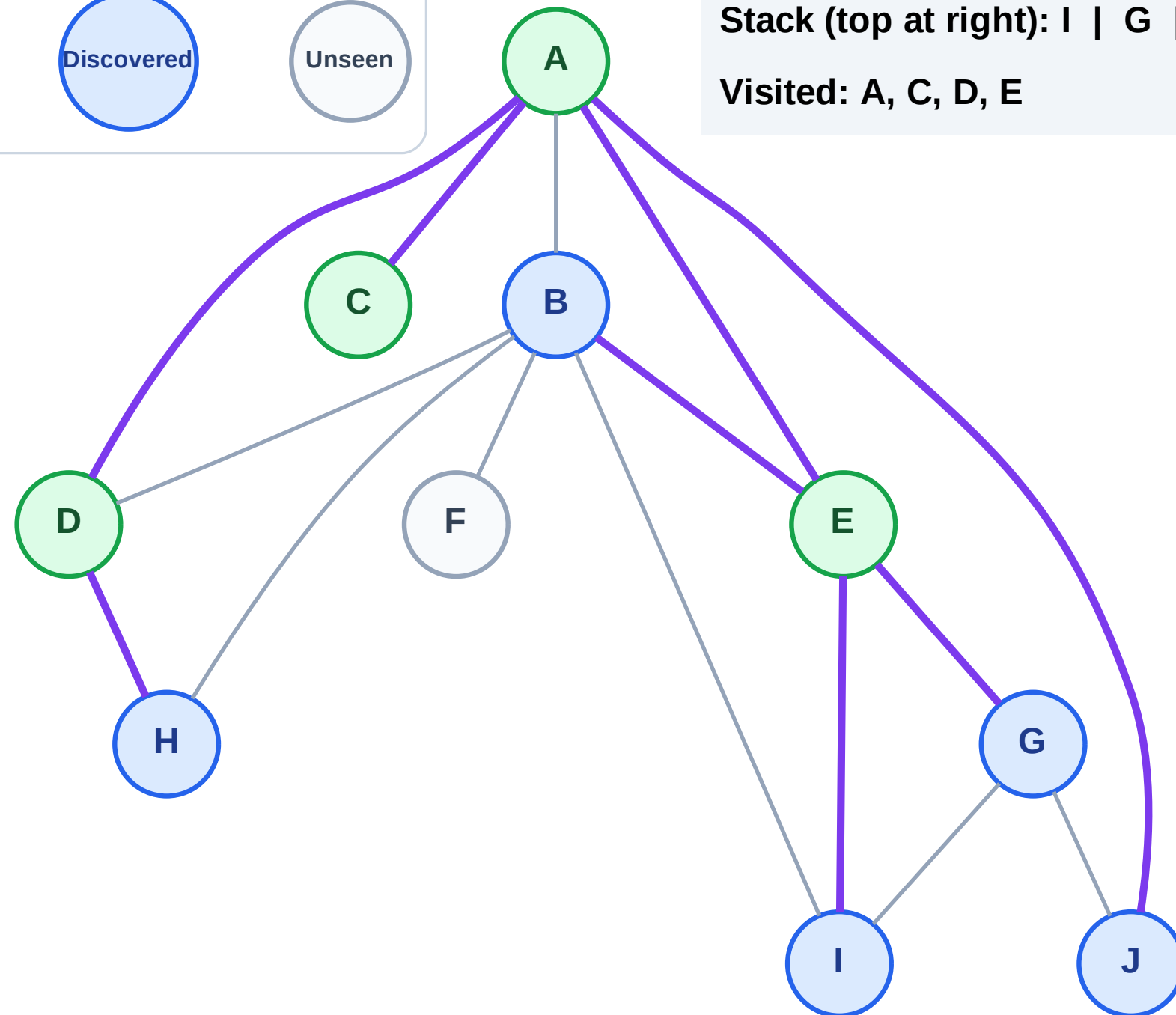
Step 9: Discover H from D

Legend



Stack (top at right): I | G | B | J | H

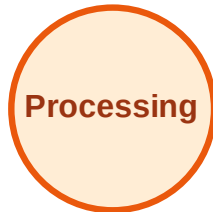
Visited: A, C, D, E



DFS Traversal

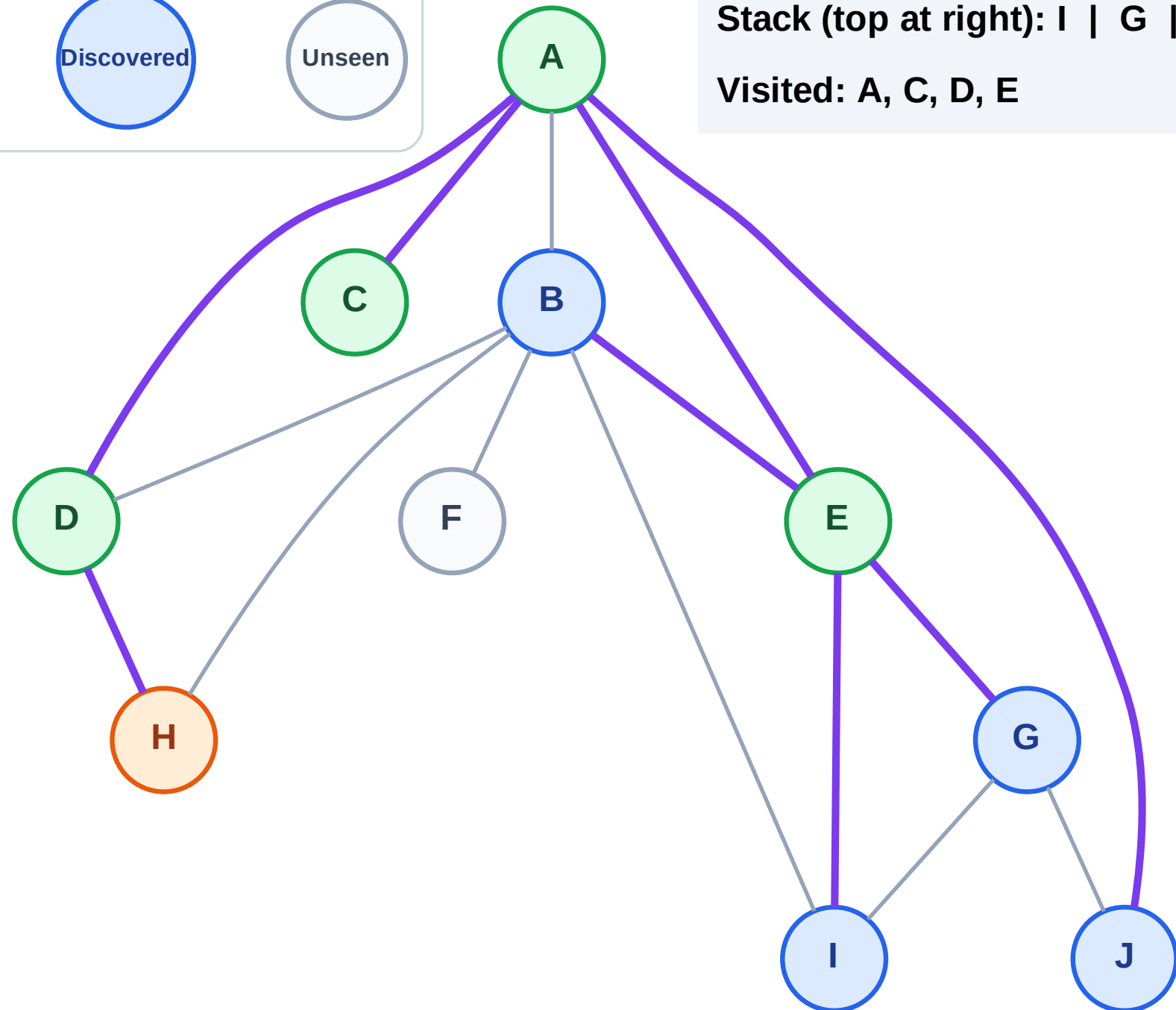
Step 10: Process node H

Legend



Stack (top at right): I | G | B | J

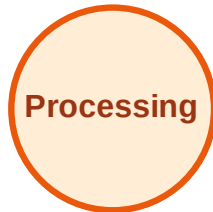
Visited: A, C, D, E



DFS Traversal

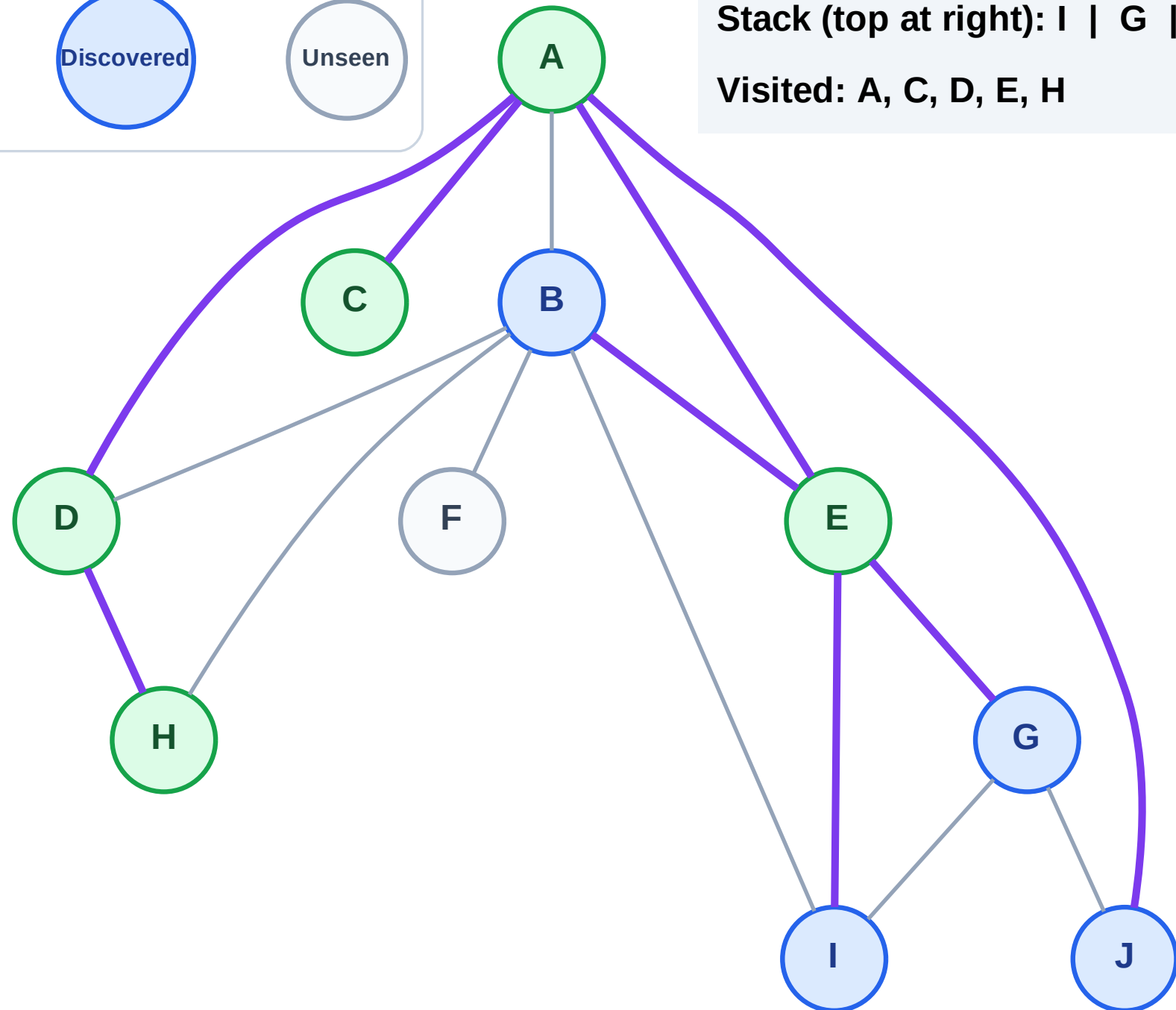
Step 11: No new neighbors from H

Legend



Stack (top at right): I | G | B | J

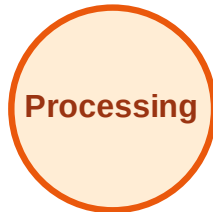
Visited: A, C, D, E, H



DFS Traversal

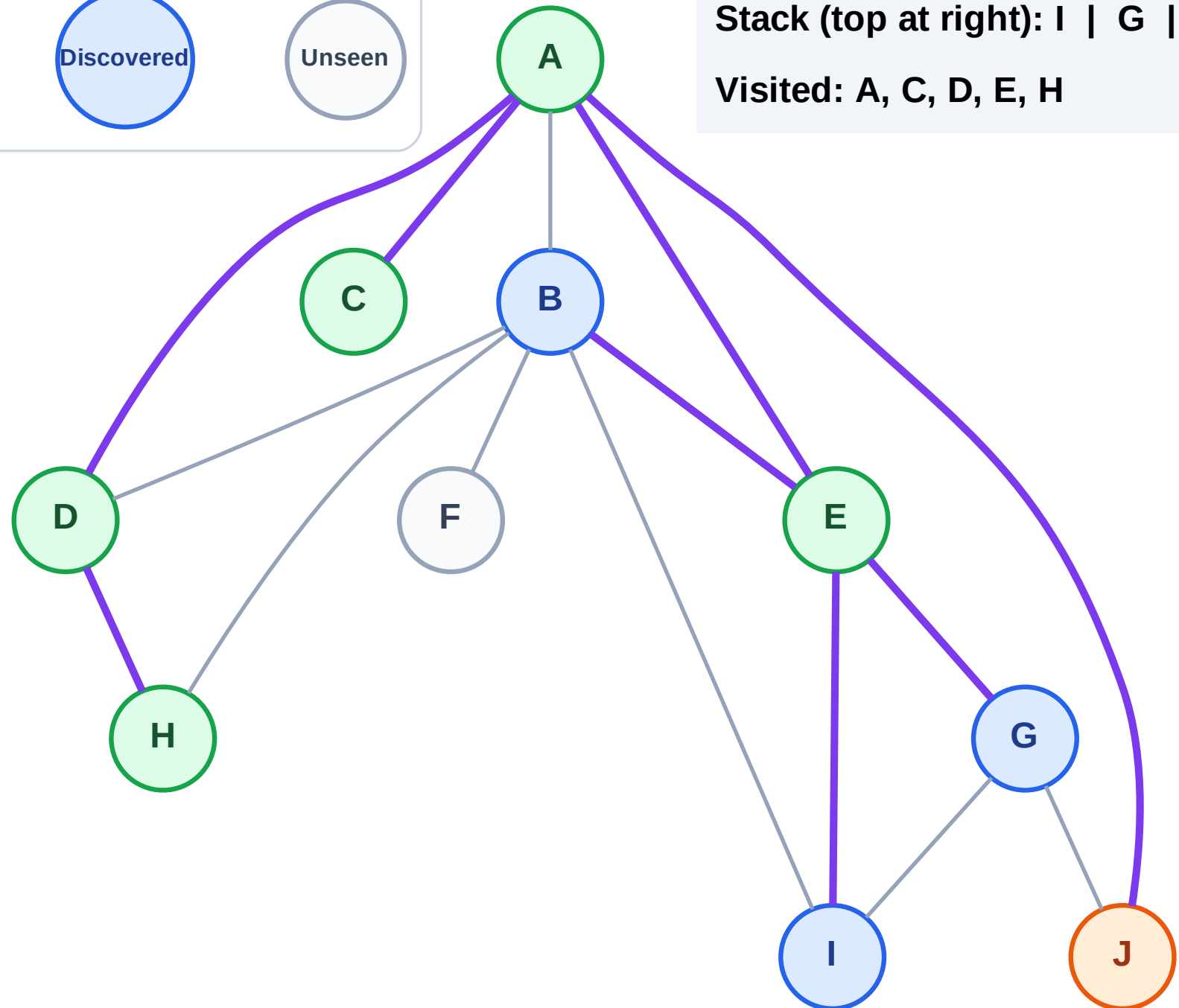
Step 12: Process node J

Legend



Stack (top at right): I | G | B

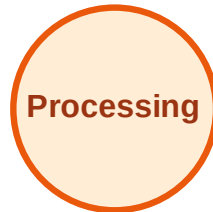
Visited: A, C, D, E, H



DFS Traversal

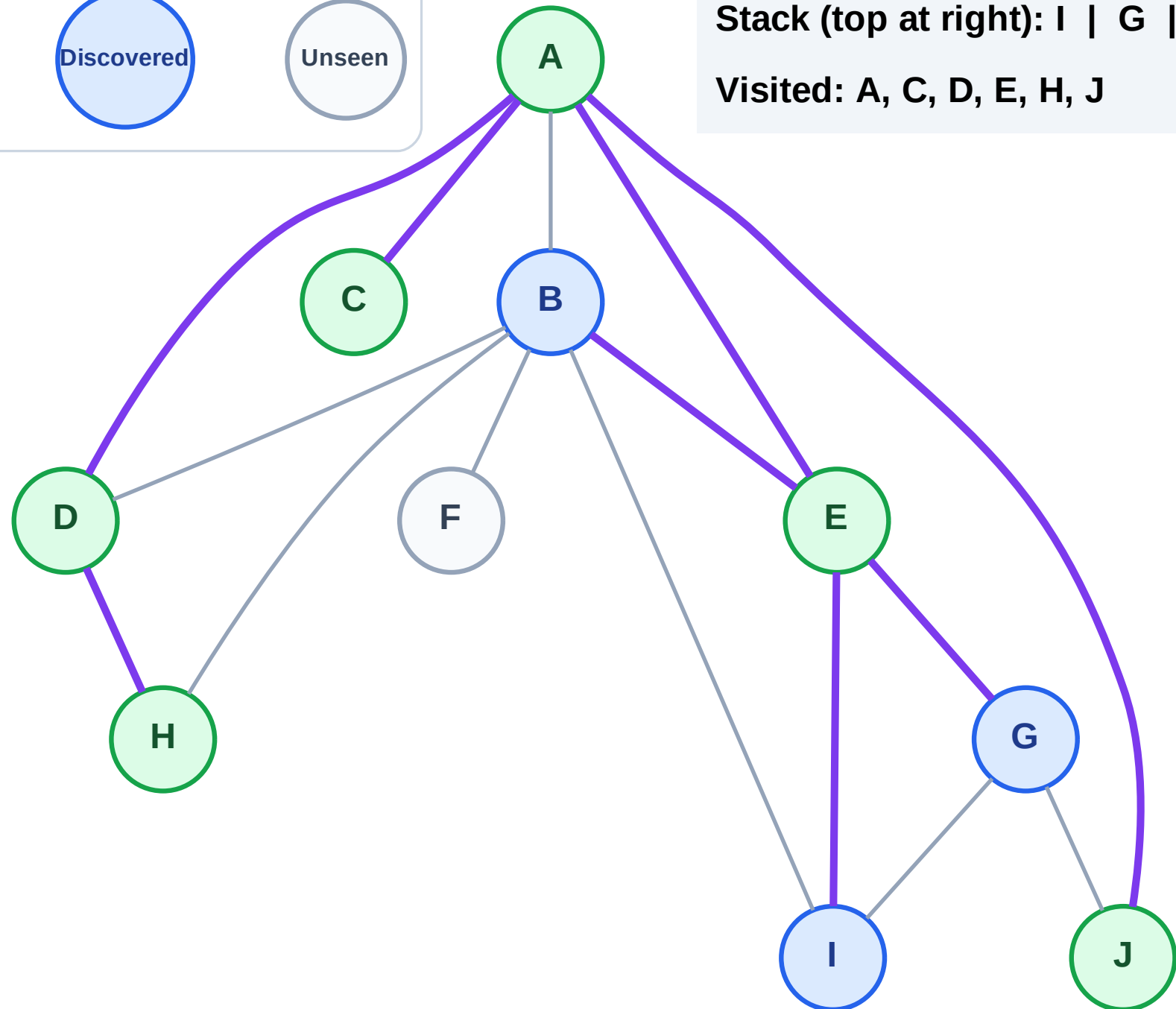
Step 13: No new neighbors from J

Legend



Stack (top at right): I | G | B

Visited: A, C, D, E, H, J



DFS Traversal

Step 14: Process node B

Legend

Complete

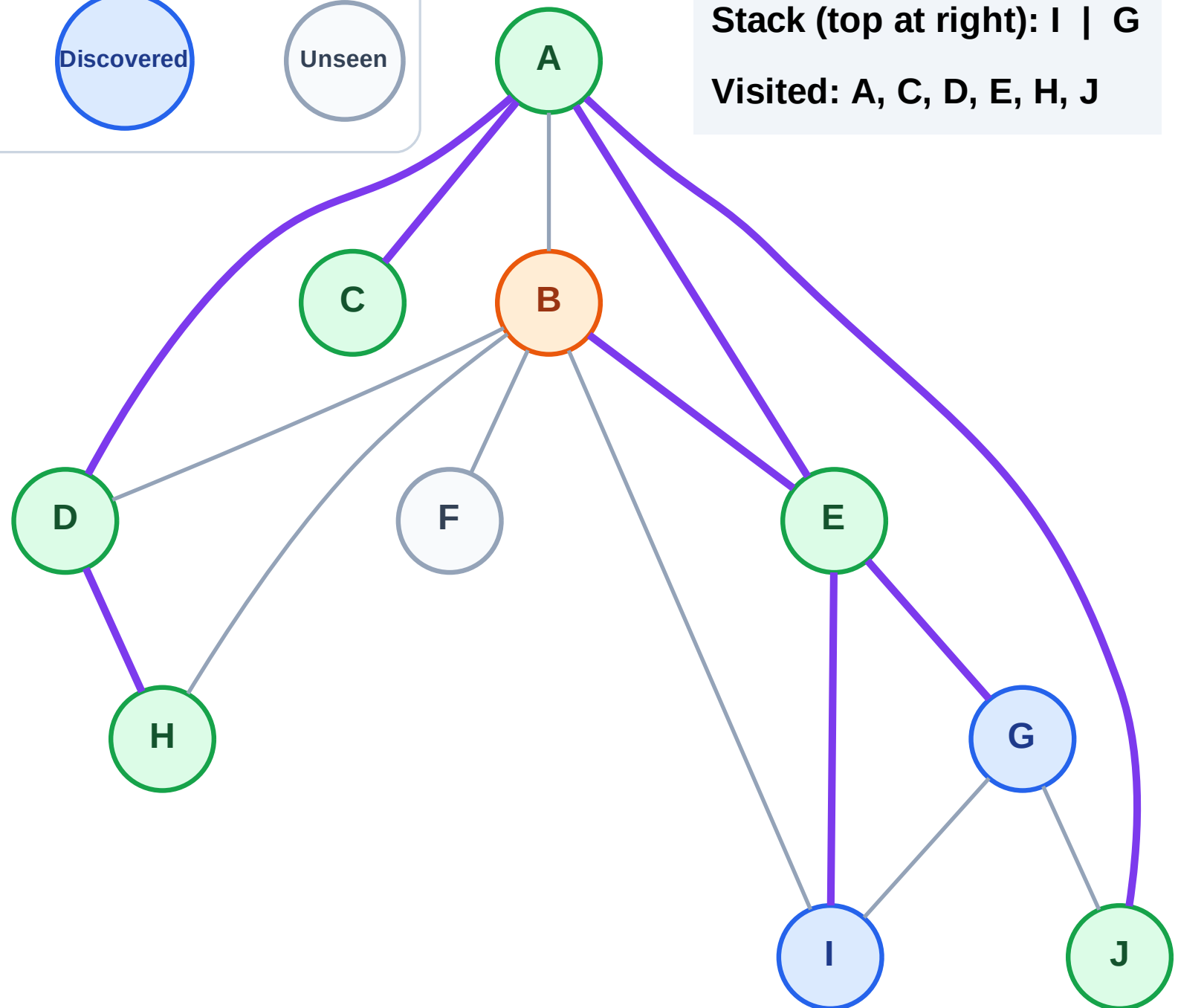
Processing

Discovered

Unseen

Stack (top at right): I | G

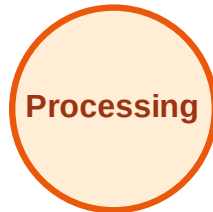
Visited: A, C, D, E, H, J



DFS Traversal

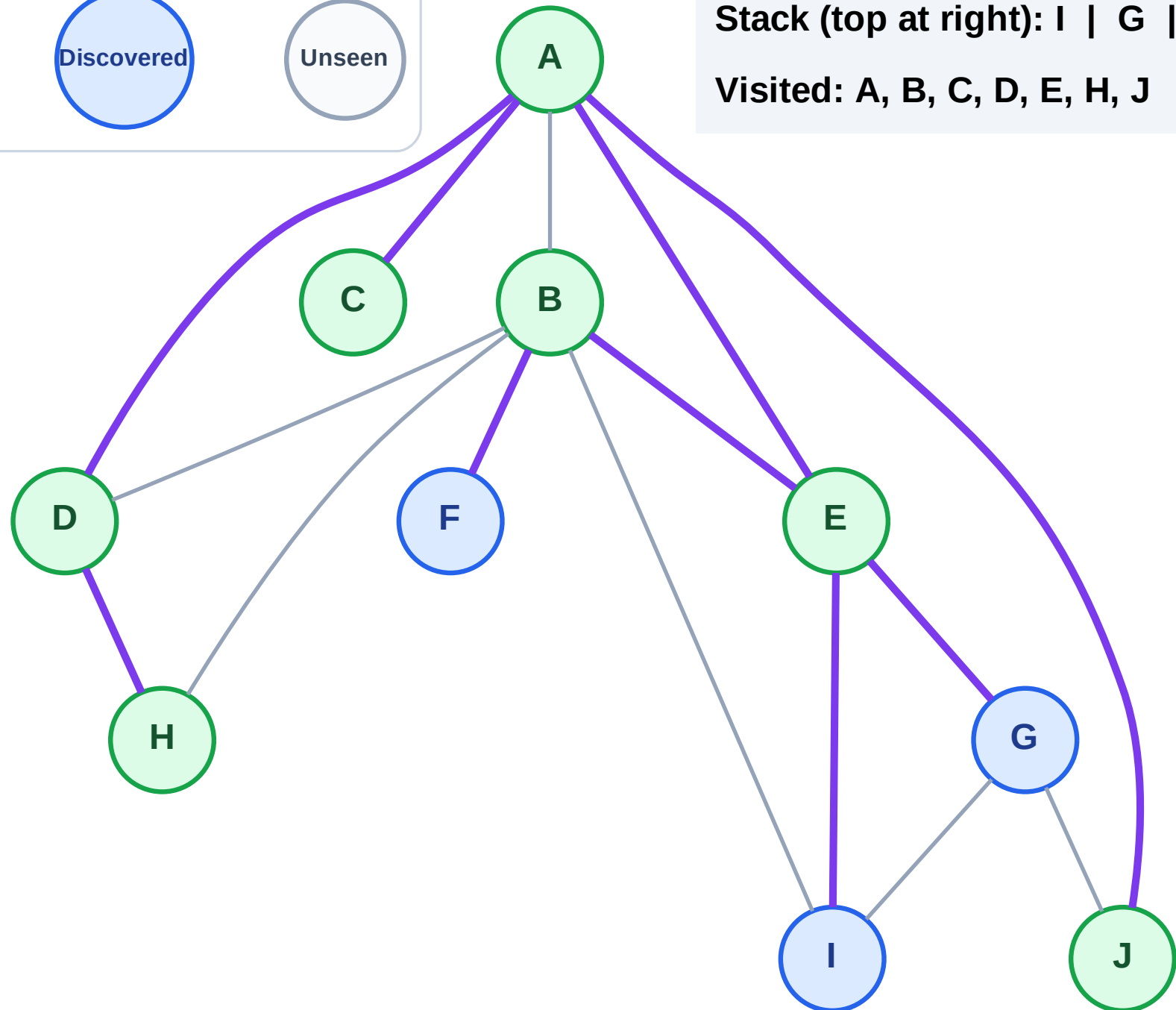
Step 15: Discover F from B

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



DFS Traversal

Step 16: Process node F

Legend

Complete

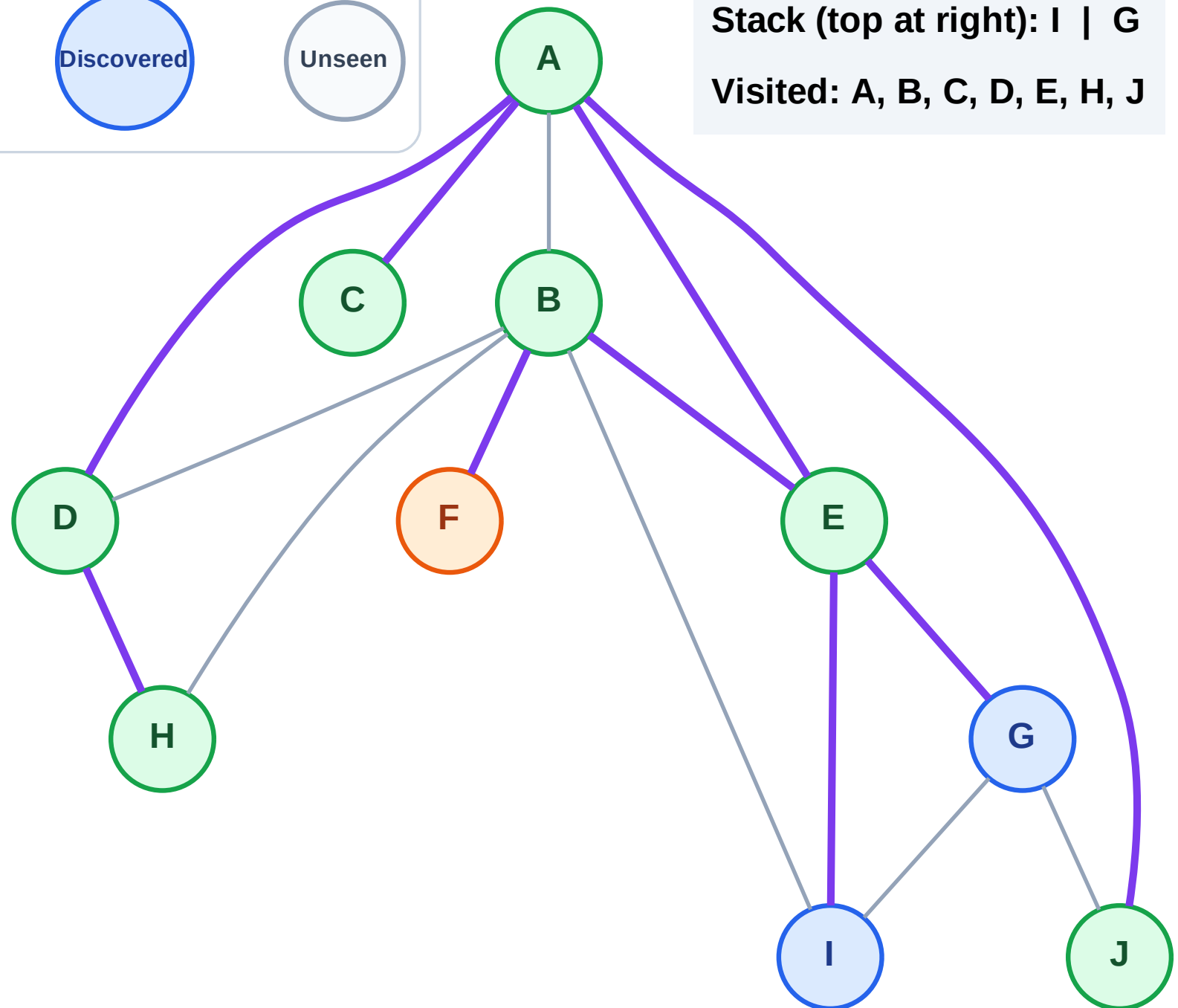
Processing

Discovered

Unseen

Stack (top at right): I | G

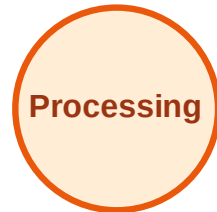
Visited: A, B, C, D, E, H, J



DFS Traversal

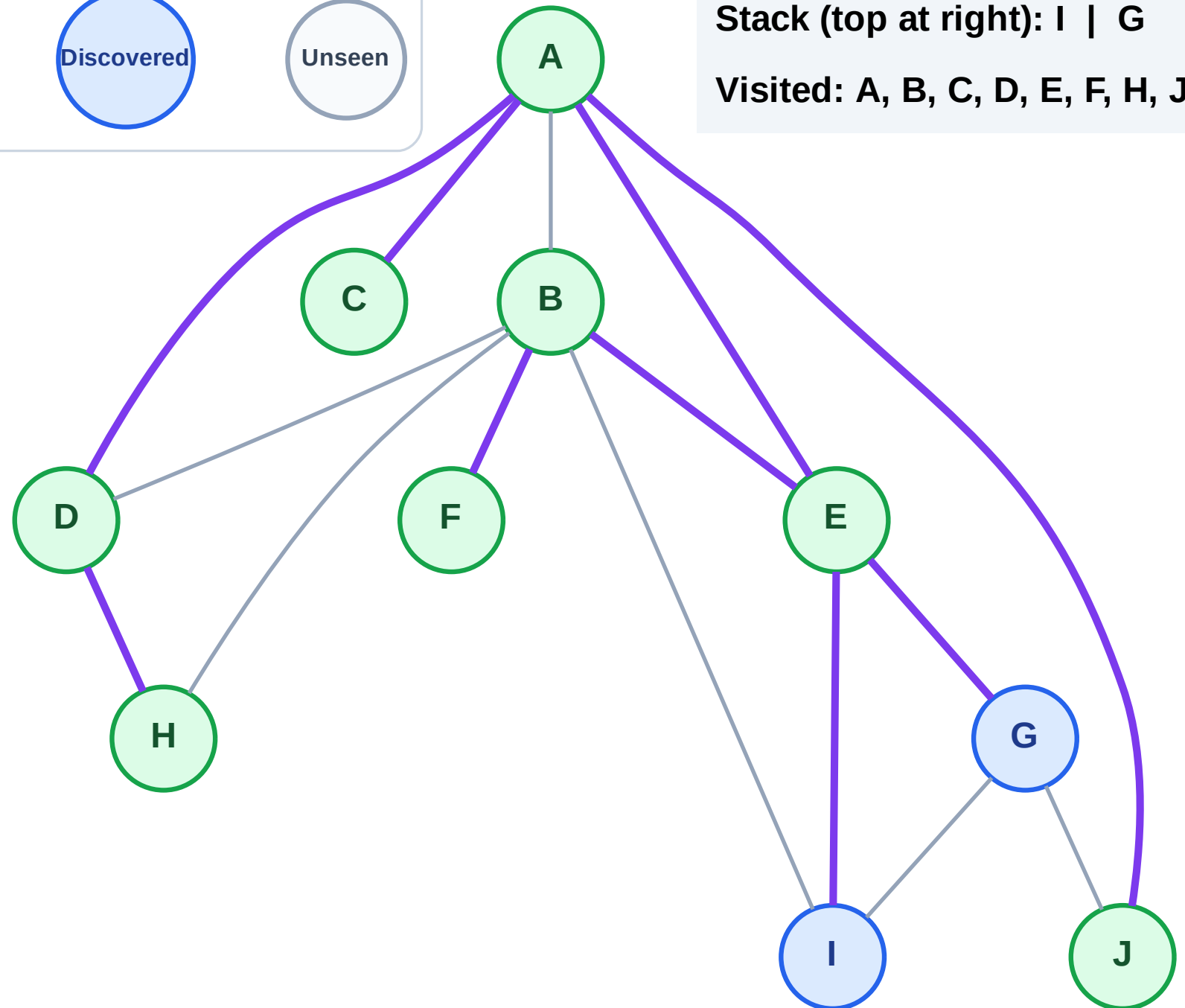
Step 17: No new neighbors from F

Legend



Stack (top at right): I | G

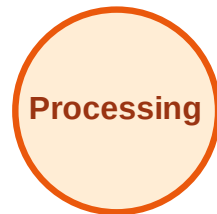
Visited: A, B, C, D, E, F, H, J



DFS Traversal

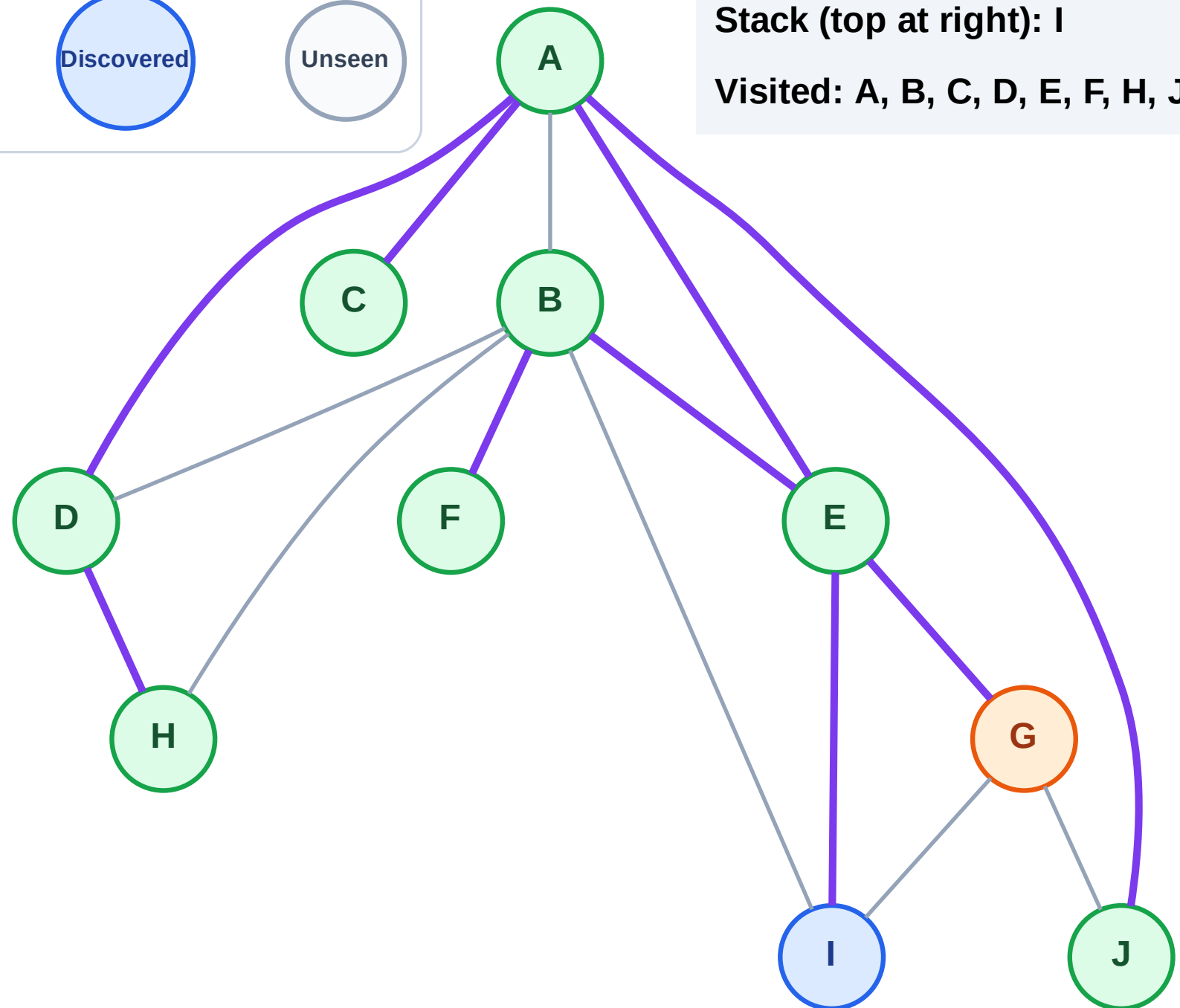
Step 18: Process node G

Legend



Stack (top at right): I

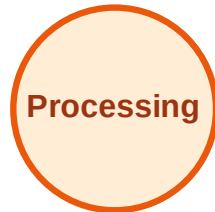
Visited: A, B, C, D, E, F, H, J



DFS Traversal

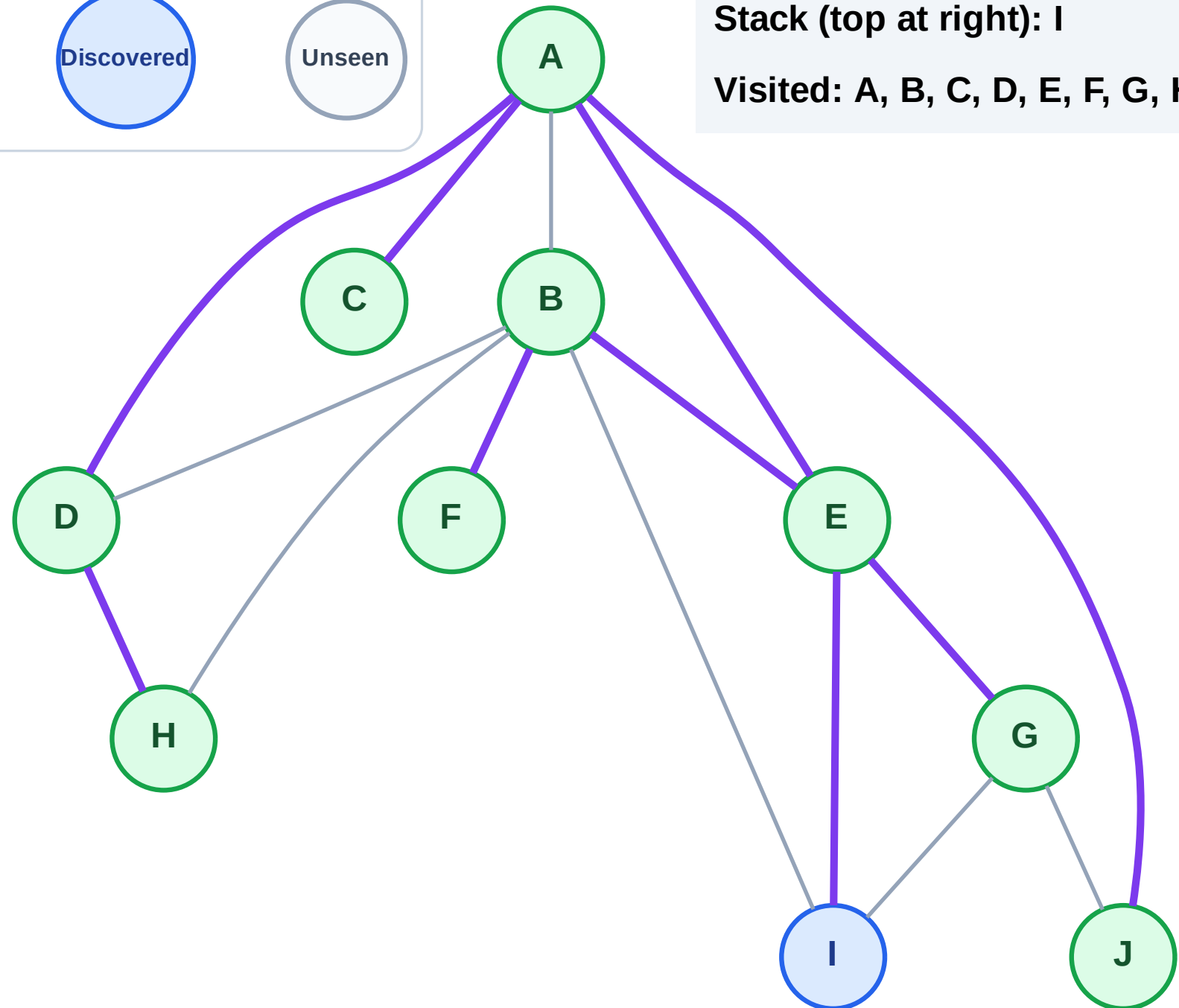
Step 19: No new neighbors from G

Legend



Stack (top at right): I

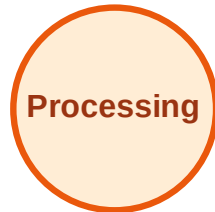
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

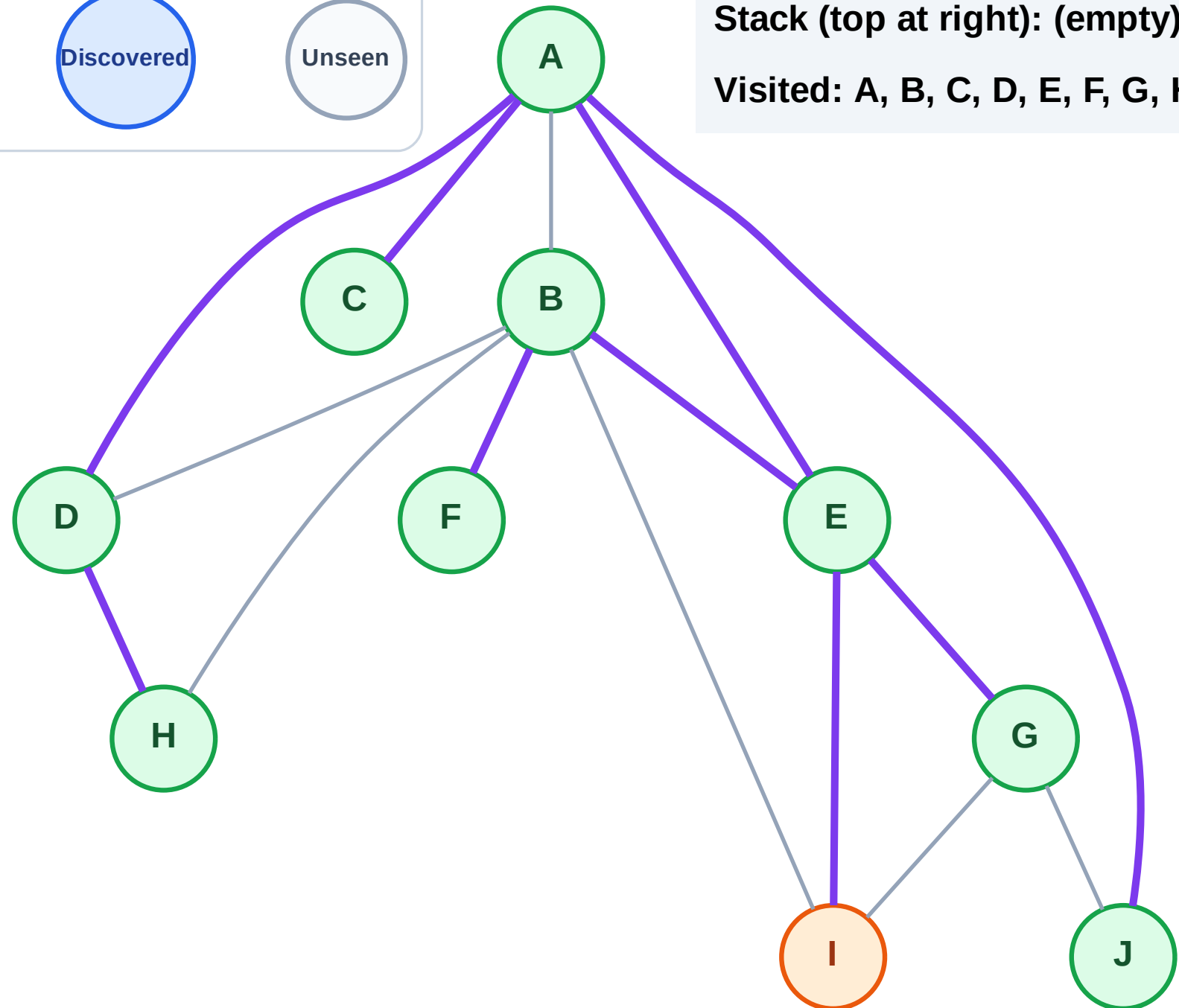
Step 20: Process node I

Legend



Stack (top at right): (empty)

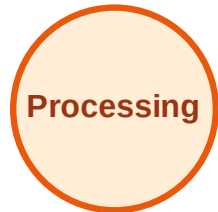
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

